Answer Part 1:

Git use and engineering workflow

TokTickIT - Lab 4 final report | 11 October 2026 | Pitchai Chadchuangchot, 67070501068. Peer: Phanuwit Butchari, 67070501070 (auto4496).

The application release passed the clean-main gate at 0f169ea. Phanuwit approved and merged the corrected evidence follow-up in PR #52, main 72314d5. The author approved the report and authorized finalization on October 11. The completion record distinguishes retained tests, agent visual checks and author report approval; it does not claim a new manual demo or screen-reader audit.

Repository | Release PR #51 | Project | Issue #44

Evidence PR #52 | Final peer approval | Published final-main evidence

Feature -> lab4-staging	Peer merge
#45 contract / ff97405	d32c8cf
#46 foundation / 4795f38	a915812
#47 action UI / 0cf9e58	f4da089
#48 workflow / b7f6bb9	00fddc1
#49 dashboards / 6dce864	4660ac6
#50 quality correction / f517961	9b5d3a5
#51 lab4-staging -> main	0f169ea; 02:34:13 Bangkok, October 11

Evidence follow-up #52: approved 27cf94d at 16:02:15 Bangkok; peer-merged as 72314d5 at 16:02:30. This documentation/checker change did not alter the fully gated application.

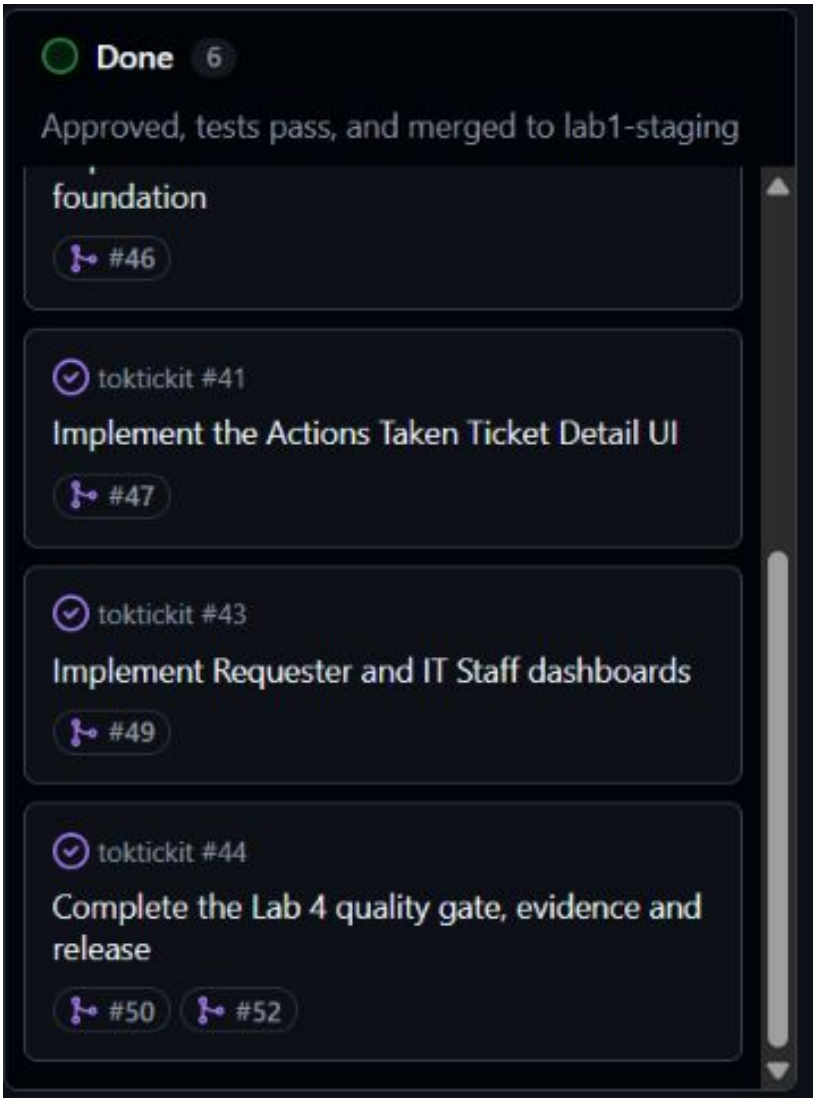
Actual issue/project snapshot

Issue	Observed card	Acceptance boundary
39	Done	Peer-accepted feature
40	Done	Peer-accepted feature
41	Done	Peer-accepted feature
42	Done	Peer-accepted feature
43	Done	Peer-accepted feature
44	Done	Author-approved closeout

Snapshot captured: 2026-10-11 09:30:57 UTC. Complete retrieved Issue/Project records and author acceptance are embedded. All six Issues are closed and their actual cards are Done.

Actual Project board - Done (6)

October 11 closeout capture. Labelled excerpt (906, 218, 1263, 700) shows Done count 6 and closed Issue #44 with merged PR #52. Full original board PNG is embedded. Column description still says lab1-staging; the actual Lab 4 release is main 0f169ea, not that historical label.



Rendered reviewer.md - selected current sections

Published reviewed source: reviewer.md. The complete maintained Markdown is embedded, including the later author-approved closeout. Published-version links and current embedded closeout text are explicitly distinguished; dated history is retained in the attachment.

Lab 4 Peer Review Record

Status: Contract and foundation accepted in PRs #45/#46. Phanuwit approved corrected Issue #41 head 0cf9e58 and merged PR #47 into lab4-staging as f4da089 on 2026-10-07. He approved Issue #42 at b7f6bb9 and merged PR #48 as 00fddc1 on 2026-10-09. He approved Issue #43 at 6dce864 and merged PR #49 as 4660ac6 on 2026-10-10. Corrected PR #50 and release PR #51 were subsequently approved and peer-merged. Clean main 0f169ea passed the final gate on 2026-10-11. Evidence PR #52 was corrected, approved and peer-merged as main 72314d5. The author then approved the report and authorized final assembly on October 11. Current Issue/Project completion is recorded separately from earlier snapshots. Reviews given by Datakung on eight Lab 4 PRs in auto4496/toktickit are verified and recorded below, separately from reviews received.

Author: Pitchai Chadchuangchot, 67070501068, GitHub Datakung. Expected peer: Phanuwit Butchari, 67070501070, GitHub auto4496. Repository: https://github.com/Datakung/toktickit

Review workflow

Six Issues were created on 2026-10-03 and linked to TokTickIT Individual Sprints: #39 contract, #40 action foundation, #41 action UI, #42 Ticket workflow, #43 dashboards and #44 quality/release. All were initially placed in Backlog. Feature PRs target lab4-staging; final reviewed release targets main. Specifications must be approved before product implementation. Normal commits/pushes are performed by the author; the peer supplies actual review comments, approval and merge.

Work group	Issue	PR/base	Review/author response/approval	State
Engineering contract	#39	#45 / lab4-staging	Corrected ff97405; actual approval below	Merged d32c8cf
Action foundation	#40	#46 / lab4-staging	Approved by auto4496 on 4795f38	Merged a915812; Issue closed/Done
Action UI	#41	#47 / lab4-staging	Corrected 0cf9e58; approval below	Merged f4da089; Issue closed
Ticket workflow	#42	#48 / lab4-staging	auto4496 approval of b7f6bb9	Merged 00fddc1
Dashboards	#43	#49 / lab4-staging	auto4496 approval of 6dce864	Merged 4660ac6; Issue closed
Quality/release	#44	#50 / lab4-staging; #51 / main; #52 / main	Actual approvals 5480214732 / 5480530699 / 5482676261	Merged 9b5d3a5 / 0f169ea / 72314d5; author report approval recorded

Actual reapproval, release and final-main verification, October 11

Phanuwit (`auto4496`) approved corrected PR #50 at exact head `f51796157dd1cc9b1c47f652213d40659f341bf5`, 01:01:29 Bangkok, and peer-merged it at 01:01:34 as `9b5d3a5454edc83dd82f29cb9a2c5f86818860b4`. He independently reran all ten gate regressions; the earlier full database-backed run remains historical, not a new run for that tree-identical correction. The author replied to the P2 finding in 4238645798.

Phanuwit approved release PR #51 at exact head `9b5d3a5`, 02:34:10 Bangkok, and peer-merged it at 02:34:13 as `0f169eaeca9bf00672c5c32884a1913c53b845d3`. He verified the accepted PRs #45-50, the unchanged corrected application tree and ten regressions; approval was not a claim that the author had already completed final-main/PDF acceptance.

The reviewer identified premature author closure/Project Done at 02:16. With explicit author authorization, #44 was reopened at 02:38:08 and its card returned to Started. Actual snapshots show #39-43 closed/Done and #44 open/Started. Do not fabricate final all-Done evidence or imply the reviewer changed the card.

The author's clean-main gate ran 02:39:09-02:45:43 Bangkok on October 11: 298 server / 211 client / 49 Chromium tests, ten gate regressions, both builds, Prisma validation, four clean audits and whitespace checks passed. No required skips/todos/flaky retries. Both expanded development digests match; the corrected manifest records `preservationVerified: true` and only then `status: passed`. Final evidence and all original per-case results are separate from the unchanged candidate record.

A separate 02:48:17-02:48:29 isolated supplementary run passed 36 metric comparisons against raw database rows and live drill-downs, bounded-list ordering, 24 healthy screen/width overflow checks and zero unexpected console/page errors. Two audit harness corrections (Category selector and expected signed-out session probes) are retained in original failure logs; no product fix or relaxed application assertion is claimed. Agent visual inspection, author acceptance and recorded reciprocal review remain distinct. Publication of this evidence follow-up and final author PDF/visual acceptance remained pending at that earlier snapshot; #44 was open/Started.

Evidence PR #52 correction, reapproval and author report acceptance

Phanuwit requested changes in review 5482569222 at head fea0905. The P2 finding showed that unchanged Windows CRLF application sources failed raw-byte hashes. The author authorized correction: compare UTF-8 CRLF-to-LF canonical actual working-file content against approved Git blobs, while retaining exact PDF, attachment, PNG and original evidence byte checks. The author responded in that review thread and pushed corrected head 27cf94d.

Phanuwit approved corrected head 27cf94d on October 11 at 16:02:15 Bangkok, with no further actionable findings. He independently passed 17 focused tests and the complete checker in clean LF and CRLF checkouts, and confirmed a real source edit failed. The PDF digest and prior visual inspection were unchanged; this was not a new full application run. He peer-merged PR #52 at 16:02:30 as `72314d5814a874d528773ba328e5d54e871ca0ef`.

The author subsequently asked whether the report was good enough. After the agent explained the remaining publication/acceptance/state updates, the author said "Go for it" to authorize finalization. Report acceptance is explicit; no new author-performed demo or screen-reader audit is inferred. The final PDF embeds current completion and actual Issue/Project snapshots, while preserving the original application gate revision and historical review records.

Reciprocal review - comments given on Phanuwit's project

Reviewer: Pitchai Chadchuangchot, 67070501068, GitHub Datakung. Peer author: Phanuwit Butchari, 67070501070, GitHub auto4496. Peer repository: auto4496/toktickit.

Recorded on October 11 from the actual GitHub review submissions, inline comments, author replies and merged-PR metadata. All PR numbers in this section refer to the peer repository, not Datakung/toktickit. The dates below use Bangkok (UTC+07:00). Test totals describe what the linked historical reviews reported; this documentation update did not rerun the peer's tests or certify his final student submission.

Reviews given and outcomes

Peer PR / scope	Datakung review and approved head	Actual outcome
#39 Contract	Changes requested: recovery test, Start control and Issue linkage. Approved 4886585, October 5, 21:34.	Merged b5bee16 into codex/lab4-staging; implementation/tests still pending at contract approval.
#41 Foundation	Approved 301c902, October 6, 14:14: preservation, repeat seed, real recovery; 391 tests and both builds reported passing.	Merged c039a7d into staging; no blocking findings.
#43 Actions	Two correction rounds: lost-response recovery, curated screenshot integrity and definitive rejection. Approved 935bcdf, October 7, 14:20; 431 tests, seven browser cases and both builds reported passing.	Merged 68a88cc into staging after corrections.
#45 Workflow	Approved a1d1fce, October 9, 15:46: atomic resolution gate, role/status rules, history and recovery; 458 tests, 14 browser cases and both builds reported passing.	Merged 69b32d6 into staging; no blocking findings.
#47 Dashboards	Changes requested: resolve generated server Prisma client without a workaround. Approved 884df39, October 10, 03:41, after normal test/capture commands passed.	Merged 72c5553 into staging after correction.
#49 Hardening	Approved fa03f754, October 10, 22:22: 484 tests, full 32-case browser suite and both builds reported passing; scale/responsive/report evidence checked.	Merged 27272813 into staging; final-main/student acceptance remained separate.
#50 Main release	Approved 27272813, October 10, 23:11: tree identical to approved fa03f754; all six increments included.	Merged 7a697d2 into main; required fresh final-main evidence, not a claimed new test run.
#51 Report polish	Approved 01a5fd6, October 11, 02:47: technical prompt summaries, Reflection, actual-main reference and report integrity.	Merged 827dca9 into main; optional source-hash suggestion was not a blocker.

GitHub identifies Datakung as the merger of all eight PRs. These peer-repository merges must not be confused with auto4496's merges of this author's own PRs. Approvals of #49-51 explicitly retained the peer's Issue #48 final acceptance boundary; they did not authorize marking the student's final submission complete.

Findings, peer responses and re-review

Contract #39: the requested-changes review identified a missing backup/restore test plan and explicit Start control, plus formal Issue #38 linkage/PR Review status. In his reply, Phanuwit identified correction 4886585: INT-02 restores records/relationships/files and checks downloads; the Start contract covers role/terminal/busy/conflict behavior with planned UI/E2E coverage; linkage/status were corrected. Datakung's linked approval confirms those documentation changes while leaving execution pending.

Actions #43: the first review reported that a committed create with a lost response and later draft edits reused the key with a different payload and stranded the draft. It also found routine tests overwriting curated PNGs without updating their manifest. Phanuwit's first reply identified 25b478c: retain/replay the original payload/key, preserve later edits, recover the saved record without duplication, and separate ignored routine captures from explicit evidence publication. Re-review confirmed both corrections, but the second finding showed definitive validation rejection still locked Save/Cancel into replaying an invalid request. In his second reply, Phanuwit identified 935bcdf: known rejection unlocks the retained corrected draft with a fresh key; ambiguous outcomes retain the original recovery request. Datakung then approved after the historical 431-test/seven-browser verification.

Dashboards #47: the requested-changes review and inline finding showed normal test/capture collection failed because the fixture imported an ungenerated root Prisma client. The initial 483-test/26-browser/both-build results were explicitly qualified: browser tests used a temporary client-path workaround. Phanuwit's reply identified 884df39: resolve Prisma through server/package.json using createRequire. Both normal dashboard commands passed four cases each without the workaround; these are the same cases executed twice, not eight distinct tests. Datakung's approval confirms the fix and consistent refreshed evidence.

Report #51: the linked approval includes an optional suggestion to record normalized committed-source hashes for reproducible supplementary checksums. It was not a requested-changes finding. No peer response or implementation of that optional suggestion is invented. Student read-through, demonstration, native-zoom review and final closeout remained separate from this documentation approval.

README, ignore rules and directory evidence

README documents setup, guarded test/E2E configuration, seed/migration instructions, builds and review ports. .gitignore excludes dependencies, secrets, builds, uploads and routine test output. Scoped exceptions retain intentional logs. PR #52 published the final-main evidence and checker correction.

Repository area	Purpose
client/src; client/tests/lab-01..04; client/e2e/lab-02..04	Role interfaces, component/style and browser regression
server/src; server/prisma/migrations; server/tests/lab-01..04	APIs, preserving schema and unit/integration/recovery tests
docs/lab-04/{specification,api-spec,ui-spec,tests,reviewer,ai-use}.md	Maintained engineering contract, traceability and actual review/AI record
artifacts/lab-04/{quality-gate,final-main,screenshots}; scripts/	Distinct historical/final evidence and reproducible guarded helpers

Answer Part 2:

Spec DD - reviewed engineering contract

The corrected contract ff97405 was approved in PR #45 and merged as d32c8cf before the foundation/product PRs #46-50. The following renders the current normative contract, not the stakeholder handout verbatim.

Rendered specification.md - selected current sections

Published reviewed source: specification.md. The complete maintained Markdown is embedded, including the later author-approved closeout. Published-version links and current embedded closeout text are explicitly distinguished; dated history is retained in the attachment.

1. Sprint goal

Complete TokTickIT with accountable Actions Taken, a backend-enforced final Ticket lifecycle and role-specific dashboards while preserving the authenticated service desk delivered in Labs 1-3.

2. Stakeholder request

Staff need to record actual work independently of the primary Ticket Owner, track follow-up and formally resolve reviewed work. Requesters need an owned summary and readable action history. Dashboards must connect to detailed screens, and the completed product must remain consistent, accessible and dependable.

3. Scope

Included: Actions Taken records/assignment/lifecycle/audit; final Ticket gate and history; Requester and Staff/Admin dashboards; preserving migrations and seed; concurrency/retry safety; responsive UI; regression and final-main evidence.

Excluded: SLA clocks/escalations, external notifications, inventory/purchasing, cost accounting/billing/payroll, multi-level approvals/signatures, advanced BI or report exports, multi-tenancy/production cloud operations, new action file uploads and optional Admin account metrics. Attachment Notes reference existing Ticket files in plain text; they do not grant Staff upload/removal permission.

4. Functional requirements

ID	Requirement	Issue
FR-01	Preserve all earlier records and authorization through an additive migration, repeatable seed and tested recovery approach.	#40
FR-02	Provide paged multiple Actions Taken under each accessible Ticket with all required fields and backend actor identity.	#40, #41
FR-03	Staff/Admin can create/edit/assign/start/complete/cancel eligible actions; Requester reads owned actions only.	#40, #41
FR-04	Reject invalid fields, ineligible assignees, stale/duplicate writes and forbidden operations without partial updates.	#40, #41
FR-05	Retain append-only action revisions and Ticket transitions in stable paged order with role-safe visibility.	#40, #42
FR-06	Enforce the final eight-status matrix and current-cycle resolution/cancellation guards on the backend.	#42
FR-07	Keep Requester resolution indication advisory and start a fresh work cycle on reopening.	#42
FR-08	Compute concise owned Requester dashboard metrics and bounded recent/attention lists.	#43
FR-09	Compute Staff/Admin operational metrics, current-user actions and bounded recent Tickets.	#43
FR-10	Supply exact URL-driven drill-down, refresh and role-specific Dashboard navigation.	#43
FR-11	Preserve all earlier auth, accounts, Tickets, files, Comments and private Notes with safe consistent feedback.	#40-44
FR-12	Verify Zen Green, keyboard/accessibility, three widths, bounded-query performance and final-main release evidence.	#41-44

5. Business rules and authorization

Normal access requires an active session, active user and completed initial password change. Existing CSRF/session/credential policies remain authoritative.

Capability	Requester	IT Staff	Administrator
Own Requester dashboard	Own Tickets	No	No
Staff dashboard/queue/detail	No	All Tickets	All Tickets
Action read and shared action history	Own Tickets	All Tickets	All Tickets
Action create/edit/assign/status	No	All eligible Tickets/actions	Same as Staff
Ticket transition/history	Owned read-only history	Read/write	Read/write
Requester resolution indication	Own eligible Tickets	No	No
Public Comments	Existing owned permission	Existing permission	Existing permission
Internal Notes	No	Read/post	Read/post
Attachment mutations	Existing owned permission	No	No
User management	No	No	Existing Admin permission

BR-01: Every action belongs to exactly one existing Ticket. No action relocation or hard deletion. Different Staff can work under the same Ticket Owner.

BR-02: Action states are PLANNED, IN_PROGRESS, COMPLETED and CANCELLED. Create starts PLANNED; transitions are PLANNED -> IN_PROGRESS/COMPLETED/CANCELLED and IN_PROGRESS -> COMPLETED/CANCELLED. No backwards/same-state transition.

BR-03: createdBy and createdAt are server-generated. performedBy and performedAt are null before completion and set to the authenticated completing actor and server time on completion. The optional assignee is independent; Staff/Admin can record work they actually performed even if assigned to someone else.

BR-04: assigneeId is null or an active IT_STAFF/ADMINISTRATOR; null means Unassigned. No requester/inactive target. Assignment of completed/cancelled work is locked. Account deactivation/change to REQUESTER unassigns PLANNED/IN_PROGRESS actions transactionally with audit events; completed historical attribution remains.

BR-05: actionAt is an explicit ISO 8601 instant with Z/offset, defaults to now in the form, cannot precede Ticket creation or exceed server now by more than five minutes. createdAt/updatedAt/performer timestamps are separate server facts. UTC is stored; Asia/Bangkok is the labeled UI display/input zone.

BR-06: description trims to 1-4000 characters; result trims to 0-4000 but completion requires nonblank result. followUpRequired is a boolean; true requires a 1-2000 character followUpNote. Optional attachmentNotes trims to 0-2000. Plain-text rendering only. Clearing a true follow-up requires an explanatory nonblank followUpNote retained in the edit/audit; do not silently erase it.

BR-07: PLANNED/IN_PROGRESS allow field edits. COMPLETED allows corrections to actionAt/description/result/follow-up/attachmentNotes only while its Ticket/cycle is active; each correction requires a 1-500 character changeReason. Completed result stays nonblank and performer/assignee/state stay immutable. CANCELLED is read-only; cancellation requires a 1-500 character cancellationReason.

BR-08: RESOLVED/CLOSED/CANCELLED Tickets reject action mutations. Only actions in the Ticket's current resolutionCycle can be changed; prior cycles stay readable. Automatic unassignment is an account-integrity exception even for legacy terminal records; it does not change Ticket status or erase actor history.

BR-09: Writes check expected Ticket version and, for existing actions, expected Action version inside the transaction. Successful action writes increment both versions/updatedAt once and append one immutable event. Lock account changes, then Ticket, then Action; recheck actor/session and assignment eligibility there. Multi-Ticket account unassignment locks Tickets/actions in ascending ID order; each affected action receives one version/event change and each affected Ticket receives one parent version/timestamp change in that account transaction. Creation includes initial assignment atomically; existing field edits and assignment use separate explicitly submitted saves, never an automatic chain. Each save is atomic on its own, not across multiple user operations. Reload observed versions after success before enabling another mutation. A later failed/uncertain

assignment cannot undo or repeat the earlier successful edit; preserve its draft and reconcile uncertainty using that operation's original key.

BR-10: Every new action mutation carries a UUID requestId. Store a durable receipt keyed by Ticket/actor/requestId with normalized operation/payload hash. Authorized exact replay returns the original receipt without another write; mismatched reuse returns 409 REQUEST_ID_REUSED. Check replay before stale version, but only after current authorization/resource access. No automatic retry with a newer version. Client reloads authoritative state after a successful receipt.

BR-11: Action events are append-only and preserve operation, actor, server time, reason and safe before/after values. Ticket events preserve transition and cycle. No public edit/delete API for either history; API transactions insert them with mutations. Order histories by createdAt ascending then id ascending. Restrictive foreign keys retain attribution. Do not put Internal Notes into shared history.

BR-12: Ticket statuses and allowed edges follow the matrix below. All transitions remain Staff/Admin-only; Requester indication never changes formal status.

BR-13: Entering RESOLVED requires at least one COMPLETED action in the current cycle, zero PLANNED/IN_PROGRESS actions in that cycle, and zero COMPLETED actions with followUpRequired=true in that cycle. CANCELLED actions never count as completed work. Check the predicate atomically while holding the parent Ticket lock. Action-list reads return an authoritative whole-current-cycle resolutionGate summary with parent version/cycle, page and counts in one repeatable-read snapshot. The UI evaluates this summary, not just loaded actions; blockers on later pages still count. Unknown/stale/mismatched summaries do not display ready or enable Resolve. A matching summary does not replace write-time gate checks.

BR-14: Entering CANCELLED requires no PLANNED/IN_PROGRESS current-cycle actions; users must explicitly complete/cancel those actions first. No completed-action requirement for Ticket cancellation. Confirm resolution/closure/cancellation in UI.

BR-15: Reopening increments resolutionCycle, clears requesterResolutionIndicatedAt and resolvedAt, and appends a transition event. Old actions cannot satisfy the new cycle gate. New successful resolution records resolvedAt=server now.

BR-16: Legacy Tickets start resolutionCycle=1, with no invented Actions Taken or audit events. Legacy RESOLVED -> CLOSED remains possible without retroactive work; any future transition into RESOLVED must meet BR-13, including a reopened Ticket. Legacy resolvedAt stays null because updatedAt does not prove the resolution time.

BR-17: Dashboard metrics come only from backend database queries in a read-only repeatable-read transaction with one asOf instant. Return nonnegative integer counts, explicit zeroes and empty arrays; never infer counts from paged client lists.

BR-18: Dashboard active Tickets are exactly NEW, OPEN, IN_PROGRESS, WAITING_FOR_REQUESTER and REOPENED. Recent means [asOf minus 7*24 hours, asOf] inclusive in UTC; UI labels this as the last seven days, not calendar weeks.

BR-19: Owned queries always include authenticated requesterId. Staff current-user queries derive actor ID from session. Never accept a dashboard identity parameter.

BR-20: Each count links to an equivalent validated list filter. Recent lists have at most five entries. Ticket lists order by updatedAt/id descending; performed action lists order by performedAt/id descending. Links use captured time boundaries so a later load does not silently redefine the dashboard range. Lists read live data: intervening writes may change their count relative to a prior snapshot; show snapshot time and offer dashboard refresh rather than promise frozen records.

BR-21: Authentication loss returns to login; forbidden/protected-not-found/conflict states remain distinct without protected details. Generic failures contain no SQL, stack, secret or file path. Preserve non-password fields on recoverable failures.

BR-22: Routine tests use guarded test/E2E databases and isolated uploads. Preserve development hashes before/after E2E; intentional screenshot capture is separate.

Final Ticket transition matrix

From	Allowed destinations	Additional gate
NEW	OPEN, CANCELLED	Cancel: BR-14
OPEN	IN_PROGRESS, WAITING_FOR_REQUESTER, RESOLVED, CANCELLED	Resolve: BR-13; Cancel: BR-14
IN_PROGRESS	WAITING_FOR_REQUESTER, RESOLVED, CANCELLED	Same
WAITING_FOR_REQUESTER	IN_PROGRESS, RESOLVED, CANCELLED	Same
RESOLVED	CLOSED, REOPENED	Reopen: BR-15; legacy close: BR-16
CLOSED	REOPENED	BR-15
REOPENED	IN_PROGRESS, WAITING_FOR_REQUESTER, RESOLVED, CANCELLED	Resolve: BR-13; Cancel: BR-14
CANCELLED	None	Terminal

All other edges fail. Existing claim/owner/priority guards, comments/notes and Requester indication policies from Lab 3 remain unless explicitly extended above.

Dashboard calculations

All Ticket metrics apply the row's current state. Refresh obtains a new snapshot.

Dashboard value	Exact predicate	Drill-down
Requester activeTickets	Owned AND status in active set	/tickets?statusGroup=active
Requester waitingForMe	Owned AND status=WAITING_FOR_REQUESTER	/tickets?status=WAITING_FOR_REQUESTER
Requester recentlyUpdated	Owned AND updatedAt in recent range	/tickets?updatedSince=...&updatedUntil=...
Requester recentlyResolved	Owned AND status=RESOLVED AND nonnull resolvedAt in range	/tickets?status=RESOLVED&resolvedSince=...&resolvedUntil=...
Requester recentTickets	Five most recently updated owned Tickets in range	Each /tickets/:id; View all uses updated range
Requester attentionTickets	Five latest owned WAITING_FOR_REQUESTER Tickets	Each detail; View all uses exact status
Staff unassignedTickets	Active AND ownerId=null	/staff/tickets?statusGroup=active&unassigned=true
Staff myTickets	Active AND ownerId=current actor	/staff/tickets?statusGroup=active&ownerId=currentId
Staff statusCounts	Group all accessible Tickets by all eight statuses, including zeroes	/staff/tickets?status=ENUM
Staff priorityCounts	Active Tickets grouped by IT Priority LOW/MEDIUM/HIGH	/staff/tickets?statusGroup=active&itPriority=ENUM
Staff myAssignedActions	Current-cycle PLANNED/IN_PROGRESS actions assigned to actor on active Tickets	/staff/actions?assignedTo=me&stateGroup=active
Staff recentPerformedActions	Five COMPLETED actions performed by actor with performedAt in range	Each /staff/tickets/:id?tab=actions&actionId=:actionId; View all uses /staff/actions?performedBy=me&state=COMPLETED&performedSince=...&performedUntil=...

Dashboard value	Exact predicate	Drill-down
Staff recentTickets	Five latest updated accessible Tickets in range	Each Staff detail; View all uses updated range

Legacy unknown resolution dates are counted in current-state totals but excluded from recentlyResolved. Explain this once in setup/data notes; never fabricate dates. Actions changes update Ticket updatedAt. Existing Comments/Notes retain their Lab 3 timestamp behavior; the metric does not promise a latest-communication timestamp.

6. UI summary

See ui-spec.md. Add /dashboard for Requester and /staff/dashboard for Staff/Admin; role homes/brand destinations use dashboards. Preserve all existing destinations and password-change routes. Shared Actions Taken lives on both Ticket detail variants; Internal Notes stay Staff-only. Dashboard links must hydrate URL filters after navigation/reload/Back, not just show filtered labels.

7. Data, migration and seed

Author-approved numbering refinement (2026-10-10): display a one-based `actionNumber` within each Ticket rather than exposing the global ID as its work sequence. The backend derives this ordinal in immutable creation order (`createdAt` ascending, `id` tie-break) across ALL of the Ticket's actions, including cancelled work and prior cycles. Pagination and assigned/performed filters do not reset it; reopening continues it. Do not derive the number from a dashboard/work-list row index. The global `id` still identifies writes, receipts, deep links and audit events. Existing records automatically receive the Ticket-local display number on read; no schema migration, stored ID rewrite, historical snapshot edit or new database index is needed for this refinement.

ActionTaken foundation: id, ticketId, cycle, state, actionAt, description, result, assigneeId?, createdById, performedById?, performedAt?, followUpRequired, followUpNote, attachmentNotes, cancellationReason?, version=1, createdAt, updatedAt. Use restrictive Ticket/User relations and bounded database string columns. Database check constraints protect positive versions/cycles, conditional follow-up notes, COMPLETED nonblank result/performer/time and CANCELLED reason. Validate the time-against-now rule in the service rather than a nonimmutable database check.

ActionTakenEvent: id, actionId, ticketId, actorId, kind, version, reason?, before/after JSON snapshots of safe action values, createdAt. Unique (actionId,version). ActionWriteReceipt: ticketId, actorId, requestId UUID, operation, payloadHash, actionId, eventId, actionVersion, ticketVersion; unique (ticketId,actorId,requestId). TicketTransitionEvent: id, ticketId, actorId, fromStatus, toStatus, cycle, ticketVersion, createdAt; unique (ticketId,ticketVersion). Ticket adds resolutionCycle=1 and nullable resolvedAt.

Use Int for IDs/versions/cycles, Boolean for followUpRequired, the documented TicketStatus/ActionState enums, UUID for requestId, timestamptz(3) for new instants, and JSONB for safe event snapshots. description/result use varchar(4000), followUpNote/attachmentNotes varchar(2000), reasons varchar(500), and payloadHash varchar(64). Before/after snapshot relationships must identify the same action and Ticket; receipts reference that action/event. Do not change legacy timestamp types as a side effect of this increment. A completed performer cannot be reassigned by an edit, even when their account is later inactive or renamed.

Indexes: actions (ticketId,cycle,createdAt,id), (assigneeId,state), (performedById,performedAt,id); events (ticketId,createdAt,id), (actionId,createdAt,id); Ticket (status,itPriority), (requesterId,resolvedAt), and (updatedAt,id) alongside preserved owner/requester indexes.

Database decisions: (1) separate current rows and immutable events reconcile editable forms with auditable append-only evidence; (2) parent-lock/version/cycle checks serialize cross-record resolution and prevent old work satisfying reopened problems; (3) nullable legacy resolvedAt avoids presenting update time as resolution time; (4) durable receipts address a lost response after a committed create.

Migration preflight snapshots schema, counts, IDs/relations, timestamps and local digests without printing credentials. Add tables/defaults/indexes without dropping earlier tables or rewriting historical ownership. Test migration from a populated Lab 3 schema, reference integrity and failure rollback in an isolated database.

Recovery: back up development before an author-run deployment. Prove restoration of a pre-migration backup into a separate disposable database and compare records/ relations. Prefer a tested forward corrective migration; never run reset/down-migration against development. Document exact recovery commands and observed results during implementation; planning alone is not passing recovery evidence.

Seed uses deterministic unique fixture keys and create-only upserts. Re-running must not reset edited users/passwords/Tickets/actions or append duplicate events. Cover all Ticket statuses/priorities, assigned/unassigned

work, zero/one/multiple actions, different Staff actors, follow-up, completed/cancelled work, and a Requester with zero Tickets. Do not rewrite existing fixtures merely to satisfy the gate.

8. API summary

See api-spec.md for exact requests, receipts, lists, queries and error precedence. Retain all existing endpoints and security; additive list filters enable dashboard links without changing existing defaults/envelopes.

Issue #44 hardening changes compatible upload/proxy dependency versions and upgrades vulnerable test tooling. It does not change the schema, action numbering, authorization, business rules or development records. Its reproducible isolated gate retains all per-case results and distinguishes release-candidate checks from the required clean-main rerun. Peer release PR #51 and evidence PR #52 are accepted and merged. Author report acceptance is recorded separately in the final completion attachment; application semantics and source are unchanged.

9. Acceptance criteria

ID	Observable acceptance criterion
AC-01	Migration/recovery and repeated seeds preserve earlier IDs, relationships, timestamps, credentials and edits.
AC-02	Multiple actions save under the correct Ticket with backend creator/performer, required fields and independent assignment. Display numbering starts at 1 per Ticket, continues across pages/cycles and agrees in detail/work/dashboard views without changing internal IDs.
AC-03	Staff/Admin create/edit/assign/start/complete/cancel only eligible work with field feedback and explicit terminal guards.
AC-04	Requester reads own actions/history only and cannot mutate actions or access Notes; anonymous/forced-change users are blocked.
AC-05	Inactive/Requester assignees are rejected; account changes preserve history and unassign active work safely.
AC-06	Stale/competing/duplicate writes have deterministic outcomes with no partial mutation within an operation, duplicate record/event or silent overwrite; independent field/assignment saves truthfully retain earlier success and reconcile later failure/uncertainty.
AC-07	Action revisions and Ticket transitions are append-only, readable past one page and stably ordered with safe visibility.
AC-08	Every Ticket matrix edge and resolution/cancellation gate is enforced through direct API requests and explanatory UI; evaluated checklist uses a version-consistent whole-cycle backend summary, including later-page blockers.
AC-09	Requester indication remains advisory; reopen clears indication, advances cycle and requires new completed work; legacy terminal records remain usable.
AC-10	Requester metrics/lists match exact owned database predicates, boundaries and zero states without cross-owner leakage.
AC-11	Staff/Admin metrics/current-user actions match predicates, contain bounded lists and return a consistent snapshot.
AC-12	Every dashboard link reaches matching filtered data, survives reload/Back and uses captured time ranges.
AC-13	Loading/validation/success/empty/forbidden/not-found/conflict/failure/retry preserve appropriate state and prevent late-response overwrite.
AC-14	Auth, all Requester flows, files, Comments, Notes, Staff functions and Administrator management retain approved regression behavior.
AC-15	Major screens are usable at 1440/768/390px with keyboard/focus, readable cues, long text and no clipping/overlap/page overflow.
AC-16	Dashboard local smoke meets documented payload/query/latency bounds and all required isolated final-main checks pass with development unchanged.

Every AC maps to planned tests in tests.md.

10. Product Definition of Done

[x] Engineering contract peer-approved before product implementation (PR #45).

[x] Every AC has passing executable evidence; no required skipped tests.

[x] Migration, preservation, repeatable seed and actual isolated recovery verified.

[x] Role/ownership, current-session, CSRF, inactive assignment, versions, retry and resolution race protection enforced directly by backend tests.

[x] Complete action/Ticket lifecycle, immutable history and legacy compatibility work.

[x] Dashboard metrics match independent queries and drill-downs match counts.

[x] Earlier functionality regressed; full tests/builds pass and production audits reviewed.

[x] Isolated E2E confirms unchanged development database/uploads.

[x] Final-main screenshots and accurate responsive/accessibility checklist inspected.

[x] Runtime console errors, broken navigation, temporary placeholder content and unfinished controls audited; setup/migration/seed/test/demo instructions verified.

[x] README and six Lab 4 docs contain real paths/results/review links/provenance.

[x] Phanuwit approves/merges features into lab4-staging and release into main; final main rerun completes and all Issues reach Done after acceptance.

[x] One readable linked PDF includes Answer Part 1-9 and author-reviewed reflection.

11. Reviewed assumptions and engineering decisions

Latest #44 verification: clean main `0f169ea` passed 298 server / 211 client / 49 Chromium cases, ten gate regressions, both builds, Prisma validation and four clean audits. Thirty-six supplementary raw-row/API/drill-down comparisons pass. This evidence follow-up does not change BR/FR/API/schema/index/numbering semantics. Evidence publication was peer-approved/merged in PR #52. Author report approval and final assembly authorization are recorded in the final completion attachment. Issue/Project closeout is based on freshly retrieved states, not an invented board.

The sheet's detailed field list does not fully specify action assignment/state, completion/cancellation or the resolution predicate, but its grading rubric asks for them. This contract proposes BR-02-16 to cover those requirements explicitly. It interprets append-only as immutable action revision/transition history, not a ban on the requested action edit UI. Assignee and actual performer are separate.

Follow-up may be cleared by a reasoned correction to a completed action while the Ticket is active; terminal Tickets and prior cycles remain immutable. Recovery does not invent legacy work. Administrator reuses Staff metrics. The seven-day window uses UTC instants and Bangkok presentation. These are project-specific choices accepted in the engineering-contract review; do not claim the handout fixed them.

Exact REST payloads, authorization/error precedence and receipts: api-spec.md. No final-evidence operation changed API or schema behavior.

Answer Part 3:

Test DD, traceability and final-main results

One uninterrupted run on clean main 0f169ea: 02:39:09-02:45:43 Bangkok, October 11. Original complete terminal logs and individual-case JSON reports are embedded in this one PDF as attachments; do not substitute earlier feature totals.

Gate step	Exit	Seconds
gate-regressions	0	1.38
development-before	0	0.73
server-tests	0	92.4
client-tests	0	35.42
server-build	0	2.6
client-build	0	4.65
prisma-validate	0	0.58
server-production-audit	0	4.37
server-full-audit	0	1.03
client-production-audit	0	0.94
client-full-audit	0	1.01
browser-tests	0	247.66
diff-check	0	0.1
development-after	0	0.35

Coverage	Actual final-main result
Server unit/API/integration/migration/recovery/regression	40 files / 298 passed; no skipped/todo
Client component/style/regression	23 files / 211 passed; no skipped/todo
Chromium end-to-end	49 passed; zero unexpected/skipped/flaky; retries=0
Gate-finalization regressions	10/10 passed
Production builds / Prisma / audits	Both builds and validation pass; four audits = zero findings
Development preservation	Identical expanded before/after fingerprint; preservationVerified=true

`SHA-256: 53a3235133832dae4df2e5e1e40711f5b844796d78b2cbdd776185e1d1d62ab3`

Complete logs are attached as server-tests.log, client-tests.log, browser-tests.log and gate-regressions.log. Complete per-case output is attached as server-results.json, client-results.json and browser-results.json. The manifest, parity record, screenshots index and complete six Markdown documents are attached too. Use a PDF viewer supporting file attachments.

Rendered tests.md - selected current sections

Published reviewed source: tests.md. The complete maintained Markdown is embedded, including the later author-approved closeout. Published-version links and current embedded closeout text are explicitly distinguished; dated history is retained in the attachment.

Lab 4 Test Plan and Traceability

Status: Contract and Issues #40-43 accepted in PRs #45-49. Issue #43 was approved at 6dce864 and merged as 4660ac6 on 2026-10-10. Corrected quality PR #50 and release PR #51 were peer-approved/merged. Clean main 0f169ea passed the complete gate on October 11, 02:39:09-02:45:43 Bangkok. Final evidence is retained separately under artifacts/lab-04/final-main/. Evidence PR #52 is approved and merged as 72314d5; the author approved the report and authorized final assembly on October 11. No new full API/browser gate or manual author demo is claimed by this closeout; historical candidate counts/provenance below are not relabelled as main evidence.

Latest final-main execution

Command: `./scripts/lab4-quality-gate.ps1 -FinalMain`. Revision: `0f169eaeca9bf00672c5c32884a1913c53b845d3`; branch main; clean at start. Ten gate-finalization regressions, 40 files/298 server tests, 23 files/211 client tests and 49 Chromium scenarios passed. Durations: server 90.85s, client 34.50s, browser command 247.66s. No required skipped/todo/flaky/retried coverage. Both production builds, Prisma validation, four zero-finding audits and diff-check pass. Original logs and individual case JSON are in `final-main/gate`; the fresh-file index distinguishes executed output from inherited candidate support files. The corrected manifest's `preservationVerified: true` precedes `status: passed`. Expanded development before/after digest: `53a3235133832dae4df2e5e1e40711f5b844796d78b2cbdd776185e1d1d62ab3`. Candidate evidence at `quality-gate/` is byte-identical to the reviewed original.

The full final-main run includes populated migration preservation, failed-DDL rollback, real disposable backup restoration, repeatable seeds, auth/CSRF/roles, files/Comments/Notes/Admin regression, all 64 matrix pairs, current-cycle/page-two blockers, synchronized races, immutable history, exact retries and Ticket-local numbering. All planned executable rows below have final-main coverage; old per-row feature counts are historical context, not the current suite's totals.

Local performance smoke (500 Tickets / 500 Actions, five warmed reads per role):

Role	Median / maximum ms	Maximum statements	Maximum JSON bytes
Requester	13 / 13	8	1697
IT Staff	13 / 17	9	3717
Administrator	12 / 13	9	2401

Separate supplementary verification at 02:48:17-02:48:29 passed 36 independently computed raw-row metric comparisons and exact live drill-down totals, four bounded list/order checks and 24 healthy screen/width audits. `dashboard-parity.json` retains query columns, predicates in the reproducible helper, actor/range/count/filter values and expected login-only 401 probes. No unexpected console/page errors or root/body overflow. The two initial harness failures and their corrections are retained; no new product fix is implied. Both additional development fingerprints match. Screenshots: 168 full-gate and 24 supplementary original PNGs, indexed with digests. Agent visual QA and author report acceptance are separate. PR #52 published the evidence; the author subsequently authorized the final report. No blanket manual screen-reader audit or new personal demo claim is made.

Planned executable coverage

Paths are repository-relative. Every executable row below passed on clean main 0f169ea on October 11. The retained individual-case reports provide exact current coverage; dated feature results below remain historical. Author report approval is separately recorded in the final completion attachment, not inferred from tests.

Test ID	AC	Scenario and expected result	Actual/planned file
MIG-01	AC-01	Upgrade populated Lab 3 schema; all historical values/relations preserved; legacy cycle/date behavior explicit	server/tests/lab-04/migration-regression.test.ts
MIG-02	AC-01	Failed migration rollback and isolated pre-migration backup restore compared against original records	server/tests/lab-04/migration-regression.test.ts
SEED-01	AC-01	Run twice after manual fixture edits; no duplicate records/events or credential/edit reset; zero/one/multiple work	server/tests/lab-04/seed-regression.test.ts
UNIT-01	AC-02,03	Description/result/time/boolean/note/reason/ID/UUID boundaries and unknown-key rejection	server/tests/lab-04/action-validation.unit.test.ts
UNIT-02	AC-08,09	All 64 Ticket status pairs and the independent current-cycle gate predicate	server/tests/lab-04/workflow-rules.unit.test.ts
UNIT-03	AC-10,11,12	UTC range boundaries, status-group/date-pair parsing and stable query rules	server/tests/lab-04/dashboard-query.unit.test.ts
API-01	AC-02,03	Multiple actions, automatic creator/performer, independent assignment, fields and valid lifecycle	server/tests/lab-04/actions-taken.api.test.ts

Test ID	AC	Scenario and expected result	Actual/planned file
API-02	AC-03,04,05	Direct forbidden/owned reads, forged actor, forced-change, inactive/Requester assignee, terminal/prior-cycle guards	server/tests/lab-04/actions-taken.api.test.ts
API-03	AC-05,06	Concurrent account change/assignment; account-integrity unassignment retains performer/events	server/tests/lab-04/action-concurrency.api.test.ts
API-04	AC-06	Simultaneous writes, stale parent/child version, exact replay after lost response, reused UUID mismatch and rollback; independent edit succeeds then assignment fails or loses response	server/tests/lab-04/action-concurrency.api.test.ts
API-05	AC-07	Revisions retained past page one; equal-time ID tie-break; correction reasons; no history edit/delete; no Notes leakage	server/tests/lab-04/action-history.api.test.ts
API-06	AC-08,09	All eight-status edges; each gate failure, legal resolution/cancellation/reopen, advisory indication, legacy terminal close; whole-cycle snapshot summary with later-page blockers	server/tests/lab-04/action-history.api.test.ts; server/tests/lab-04/ticket-workflow.api.test.ts
API-07	AC-06,08,09	Action update/completion/creation races against resolve/cancel/reopen; one consistent outcome	server/tests/lab-04/workflow-concurrency.api.test.ts
API-08	AC-10	Independent owned queries, zero account, other-owner exclusion, exact inclusive UTC boundaries and legacy null dates	server/tests/lab-04/requester-dashboard.api.test.ts
API-09	AC-11	Staff/Admin metrics, current-user work, zero status/priority keys, snapshots and <=5 lists match queries	server/tests/lab-04/staff-dashboard.api.test.ts
API-10	AC-12	Additive Ticket/work-list filters match each dashboard predicate and retain existing defaults/errors	server/tests/lab-04/dashboard-drill-down.api.test.ts; dashboard-query.unit.test.ts; requester-dashboard.api.test.ts; staff-dashboard.api.test.ts
API-11	AC-04,13	Session/CSRF restrictions, protected resources and generic errors; service remains healthy after failure	server/tests/lab-04/actions-taken.api.test.ts; requester-dashboard.api.test.ts; staff-dashboard.api.test.ts; dashboard-drill-down.api.test.ts; safe-errors.api.test.ts
HARD-01	AC-13,14,16	Aborted multipart before async path assignment leaves no orphan; service stays healthy; default proxy trust is off and mapped unrelated addresses stay untrusted	server/tests/lab-04/upload-hardening.test.ts; proxy-trust-hardening.test.ts
GATE-01	AC-14,16	Only valid matching final fingerprints permit passed; external snapshot/log-read/invalid/missing/mismatched fingerprints fail; earlier failures/exit codes persist	scripts/tests/lab4-quality-gate.test.ps1
API-12	AC-02,04,06,07,11,12	Interleaved Ticket IDs, tied creation times, page 2, prior/cancelled work, filtered work/dashboard parity, preserved write/replay/audit IDs and forged-number rejection	server/tests/lab-04/action-numbering.api.test.ts
UI-01	AC-02,03,05,06,13	View-first/Edit/Close detail, direct Start/Complete/Cancel, return focus/discard, late-history protection, independent field/assignment/state saves, required feedback, input retention, exact retry key, earlier-success/later-failure feedback and saved-but-refresh-failed state	client/tests/lab-04/ActionsTaken.test.tsx

Test ID	AC	Scenario and expected result	Actual/planned file
UI-02	AC-04,07,08,09,13	Gate/history/reopen/read-only view, paging, advisory indication and safe conflict reload; labelled checklist colours exclude stale/unknown positive results; separate opening/progress stages and explicit target retain all valid transitions	client/tests/lab-04/TicketWorkflow.test.tsx
UI-03	AC-10,12,13	Owned dashboard zero/loading/ready/failure/refresh, labels and real drill-down navigation	client/tests/lab-04/RequesterDashboard.test.tsx
UI-04	AC-11,12,13	Staff/Admin metrics/work/safe states, meaningful zeroes and bounded lists	client/tests/lab-04/StaffDashboard.test.tsx
UI-05	AC-12,13	URL hydration, invalid query, reload/Back and stale responses after navigation	client/tests/lab-04/DashboardNavigation.test.tsx; dashboards.spec.ts browser
UI-06	AC-04,07,13,15	Changed-only audit fields, initial creation details, Yes/No, Bangkok dates, honest account IDs, escaped text and incomplete legacy snapshots without mutation	client/tests/lab-04/ActionHistoryChanges.test.tsx
STYLE-01	AC-15	Reused tokens, labeled states and responsive rules where meaningful computed-style checks apply	client/tests/lab-04/ui-style.test.tsx; client/e2e/lab-04/dashboards.spec.ts
REG-01	AC-14	Real-session auth and earlier owned/file/Comment/Note/Staff/Admin flows with the approved new gate	server/tests/lab-01, lab-02 and lab-03 (existing executable files); dedicated final release coverage remains #44
PERF-01	AC-11,16	Bounded query count/payload and measured dashboard smoke on representative seeded data	server/tests/lab-04/dashboard-performance.api.test.ts
E2E-01	AC-02-07,13,15	Different actors create/assign/edit/start/complete/cancel multiple actions; independent edit then failed/uncertain assignment; Requester read-only and direct denial	client/e2e/lab-04/actions-taken-flow.spec.ts
E2E-02	AC-07-09,13,15	Missing/active/follow-up work including later-page blockers prevents resolution/checklist readiness; corrected work resolves/closes; reopen needs new cycle; status stages, checklist colours/contrast, view-first/Edit and readable audit with exact original snapshots at three widths	client/e2e/lab-04/ticket-resolution.spec.ts
E2E-03	AC-10-13,15	Both dashboards, current-user work, query parity, exact drill-down and zero/failure states	client/e2e/lab-04/dashboards.spec.ts
E2E-04	AC-14,16	Authentication/change/logout, Requester/file flows, communication/privacy and Admin safeguards remain valid	client/e2e/lab-02/requester-ticket-flow.spec.ts; lab-03/auth-review.spec.ts; admin-users.spec.ts; staff-ticket-operations.spec.ts; release-evidence.spec.ts, included by test:e2e:review
EVID-01	AC-10-16	Deliberate final-main captures at three widths, query comparison and unchanged development state	scripts/lab4-quality-gate.ps1; client/e2e/lab-03/release-evidence.spec.ts; client/e2e/lab-04/dashboards.spec.ts; client/e2e/lab-04/ticket-resolution.spec.ts

Answer Part 4:

AI use and My Reflection

Rendered ai-use.md - selected current sections

Published reviewed source: ai-use.md. The complete maintained Markdown is embedded, including the later author-approved closeout. Published-version links and current embedded closeout text are explicitly distinguished; dated history is retained in the attachment.

Lab 4 AI Use and Reflection

Status: Actual interaction record through peer-accepted Issues #40-43 and the Issue #44 release-candidate work on 2026-10-10 and author-authorized final-main verification on October 11. PR #50/#51 are peer-approved/merged and the complete main gate passed. PR #52 published the approved evidence as main 72314d5. After asking whether the report was good enough, the author replied "Go for it" to authorize its finalization. This records report approval, not an invented new manual demo, model selection history or accessibility certification.

Tool: OpenAI Codex desktop. On 2026-10-10 the author reported the selected model as "GPT-Sol 6.1". This is the author's selection record, not an independently verified model history for every earlier turn; no historical model changes are inferred from the app name or prior labs.

Selected real prompts so far

#	Prompt	Use of the result
1	"New lab here." with SE+Lab+4.pdf	Read the full sheet, inspected its illustrations/rubric and compared it with the reviewed Lab 3 schema and workflow.
2	"Ok" after the six-Issue proposal and drafting next step	Authorized local Issue/contract drafts. Proposed explicit action lifecycle, performer/assignee distinction, append-only revisions, current-cycle resolution gate, preserving migration and exact dashboard queries before implementation.
3	"Good make sure it goes with the PDF I gave you and cover some cases"	Rechecked the complete labsheet and grading requirements, added CASE-01-25 and clarified action ordering, live dashboard drill-down, database types and retry/actor isolation.
4	"Okay start"	Checked for duplicates, created Issues #39-44 on GitHub and added them to the project Backlog; prepared local staging/Issue #39 branches and moved the contract into the repository before product coding.
5	"Yeah" after the agent explained Phanuwit's PR #45 findings and asked to update both	Authorized contract corrections: separate field/assignment saves with partial-success and exact-retry behavior, whole-cycle backend checklist summary, CASE-26/27 and factual review recording. Phanuwit later approved ff97405 and merged PR #45 before product implementation.
6	"Docker ready" with the prepared feature/40-actions-foundation branch output	Authorized the next Issue's implementation/testing. Added preserving schema/migration, Action APIs, immutable history, retry receipts, safe account unassignment and repeatable demo fixtures; verified migration rollback/real isolated restore, concurrent writes and full regression. Fixed issues exposed by tests, without migrating/seeding development.
7	"On to the next" after PR #46 was reviewed/merged	Started Issue #41 from the peer-merged staging commit. Wrote failing component tests before the shared Action screen; implemented separate saves, confirmations, read-only history and exact retries. Real-browser tests exposed timestamp precision, stale completed Result and Requester deep-link bugs; visual inspection exposed a mobile caption issue. Corrected these and reran tests/evidence without changing development data.
8	"Yes fix them" after Phanuwit's PR #47 Changes requested review	Added typed Action/assignee mocks in legacy detail fixtures and a test-only guard that exposed unexpected real fetches before the fix. Kept existing cases/assertions, synchronized mutation summary versions and ensured the navigation race actually sends a claim. Reran all 130 client tests in default and serial modes and the typed client build; no product/database change or peer approval claimed.
9	"He merged it and ready to move on to next issue" with PR #47	Verified approval of corrected 0cf9e58 and merge f4da089, then started #42 from staging. Wrote failing matrix/gate/API and workflow component tests; implemented atomic current-cycle gates, transition history, reopen cycles and server-summary checklist. Added synchronized races, rollback and real-browser later-page/reopen/unknown-response checks. Kept development unchanged.
10	"no for attachment fail I think show fail but not created" after "Do it" authorized creation redirect	Implemented redirect only after initial uploads succeed; attachment failure stays on the form with a failure alert and truthful saved-Ticket retry link. Added deferred-upload/unmount and real-browser success/failure/retry checks. Isolated the older login fixture from real Staff API reads without weakening assertions. The author separately backed up and migrated development; these tests preserved it.

The final submission requires 6-10 selected actual key prompts. Ten are recorded so far. Do not invent future prompts; replace a less representative selection if later review/testing interactions warrant inclusion.

Specification-agent and coding-agent roles

Specification support transformed the supplied sheet into numbered requirements/rules/criteria, role and transition matrices, database decisions and test plans before product coding. Coding support implemented the approved behavior and regressions, responded to real peer findings, and retained truthful failed/passing runs. Human input shaped UI size, clarity, equal controls, audit presentation and Ticket-local numbering. AI-generated work was checked through tests, peer review and author UI feedback; passing automation is not invented author visual acceptance.

My Reflection - supplied by the author

AI helped me most by listing the details I needed for planning. I could also suggest my own ideas and incorporate them into the plan to make it more suitable. I still needed to check UI details myself, such as whether the buttons and page sizes were appropriate. Most of the checks I needed to make personally were on the frontend.

The author reported the selected model as GPT-Sol 6.1. This records the supplied selection, not a reconstructed or independently verified model history for every turn. The reflection is grammar-edited only. The ten selected prompts above are actual recorded interactions; later release interactions remain supplementary.

Answer Part 5:

Working IT Staff dashboard

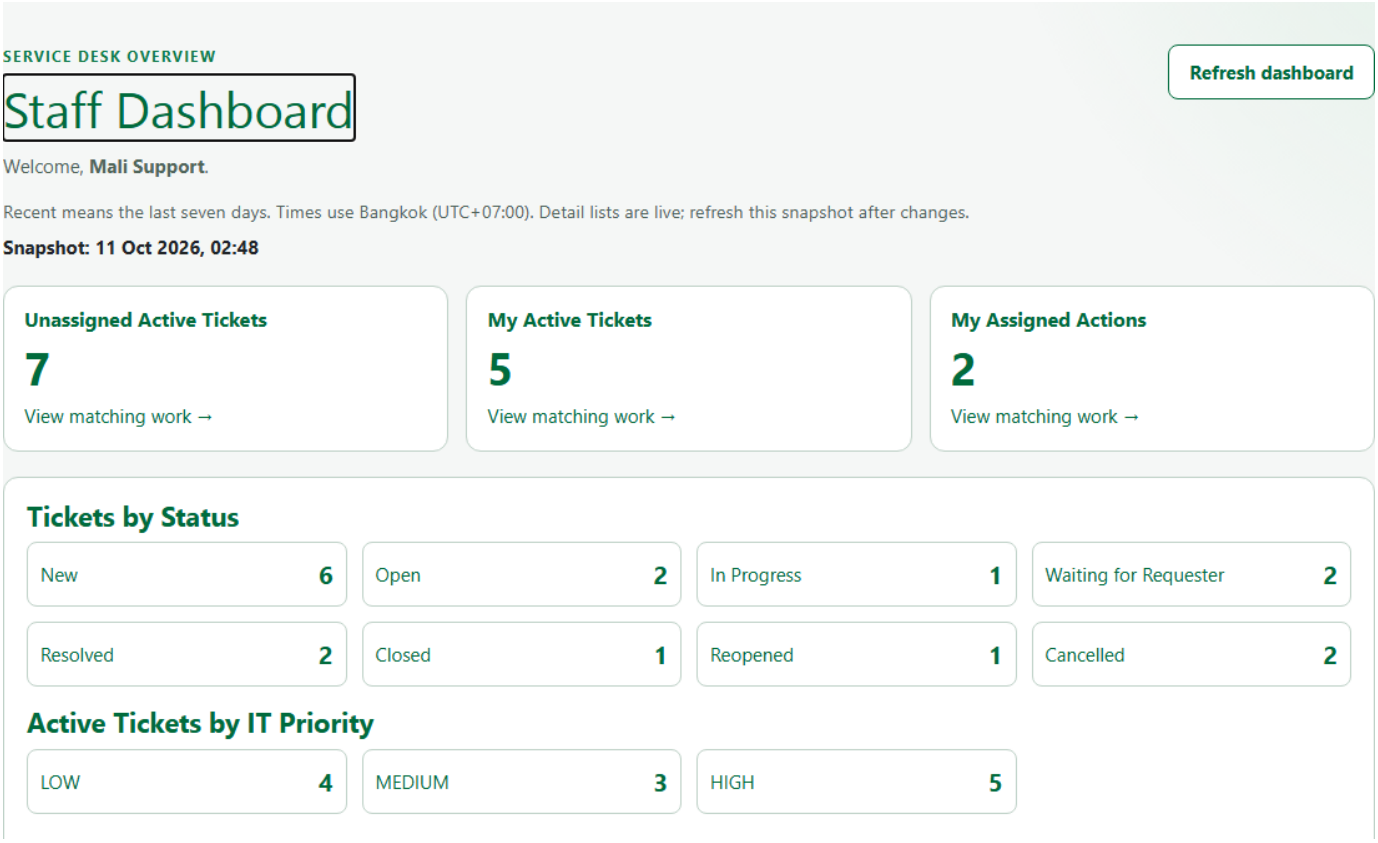
Backend repeatable-read snapshots provide active unassigned/owned work, all eight statuses, active priorities, current-user assigned actions and bounded recent/performed lists. Admin reuses Staff behavior; My counts use the signed-in actor, not another staff member. Detail lists read live data using exact captured filters. Initial loading is not fake zero; refresh failure labels the retained snapshot and offers Retry.

Actor / metric	API	Independent rows	Live list
IT_STAFF / unassignedTickets	7	7	7
IT_STAFF / myTickets	5	5	5
IT_STAFF / myAssignedActions	2	2	2
IT_STAFF / statusCounts.NEW	6	6	6
IT_STAFF / statusCounts.OPEN	2	2	2
IT_STAFF / statusCounts.IN_PROGRESS	1	1	1
IT_STAFF / statusCounts.WAITING_FOR_REQUESTER	2	2	2
IT_STAFF / statusCounts.RESOLVED	2	2	2
IT_STAFF / statusCounts.CLOSED	1	1	1
IT_STAFF / statusCounts.REOPENED	1	1	1
IT_STAFF / statusCounts.CANCELLED	2	2	2
IT_STAFF / priorityCounts.LOW	4	4	4
IT_STAFF / priorityCounts.MEDIUM	3	3	3
IT_STAFF / priorityCounts.HIGH	5	5	5
ADMINISTRATOR / unassignedTickets	7	7	7
ADMINISTRATOR / myTickets	0	0	0
ADMINISTRATOR / myAssignedActions	0	0	0
ADMINISTRATOR / statusCounts.NEW	6	6	6
ADMINISTRATOR / statusCounts.OPEN	2	2	2
ADMINISTRATOR / statusCounts.IN_PROGRESS	1	1	1
ADMINISTRATOR / statusCounts.WAITING_FOR_REQUESTER	2	2	2
ADMINISTRATOR / statusCounts.RESOLVED	2	2	2
ADMINISTRATOR / statusCounts.CLOSED	1	1	1
ADMINISTRATOR / statusCounts.REOPENED	1	1	1
ADMINISTRATOR / statusCounts.CANCELLED	2	2	2
ADMINISTRATOR / priorityCounts.LOW	4	4	4
ADMINISTRATOR / priorityCounts.MEDIUM	3	3	3
ADMINISTRATOR / priorityCounts.HIGH	5	5	5

Raw query columns and independent row predicates, actor IDs, captured ranges, filters and list ordering are embedded in dashboard-parity.json. The supplementary screenshots below use that exact fixture population; the mutable earlier full-suite screenshots are not mixed with these counts.

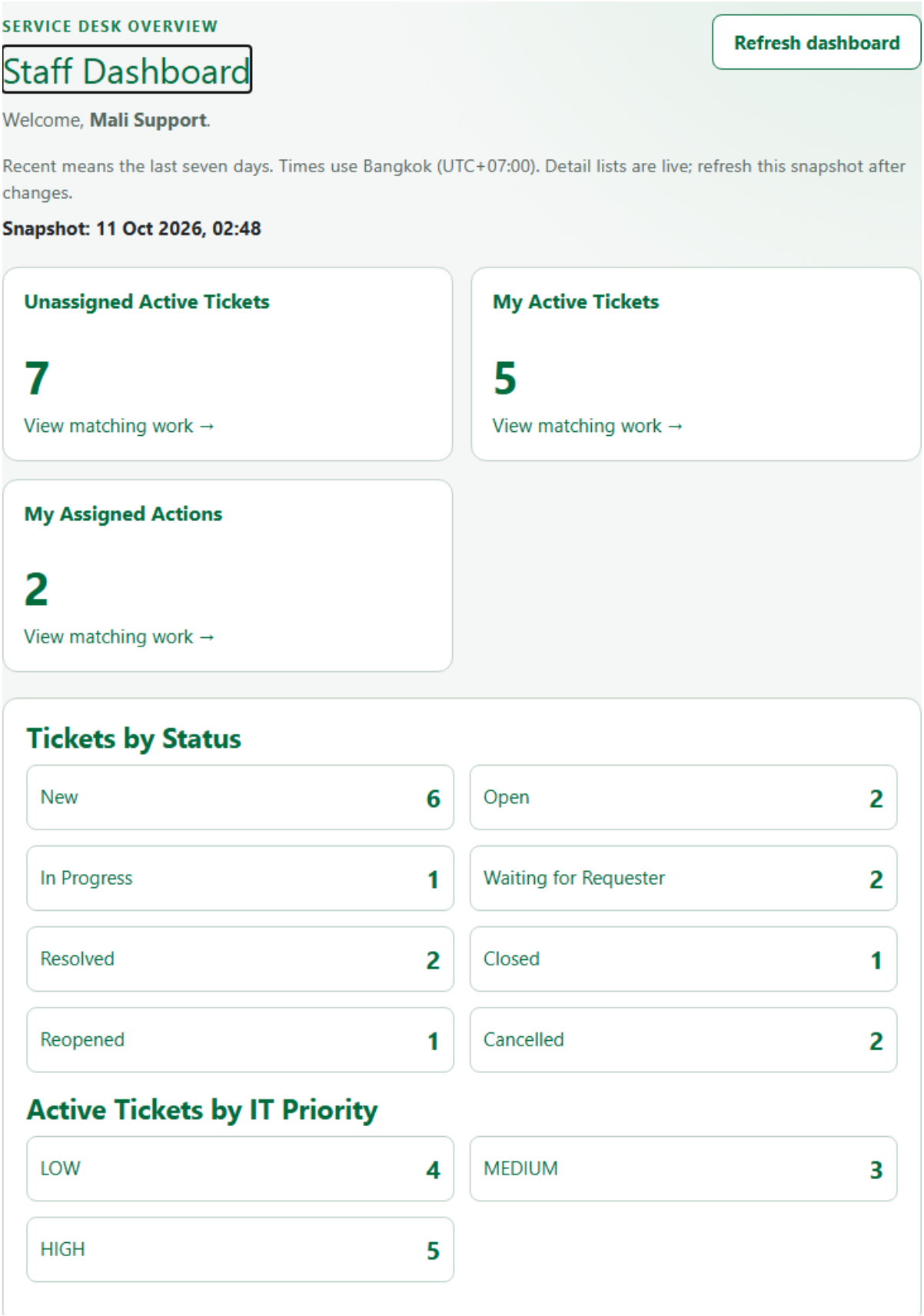
Operational snapshot - desktop

Figure 1 | staff-dashboard-1440.png | application 0f169ea. Excerpt: (170, 180, 1270, 850). Original full PNG retained.



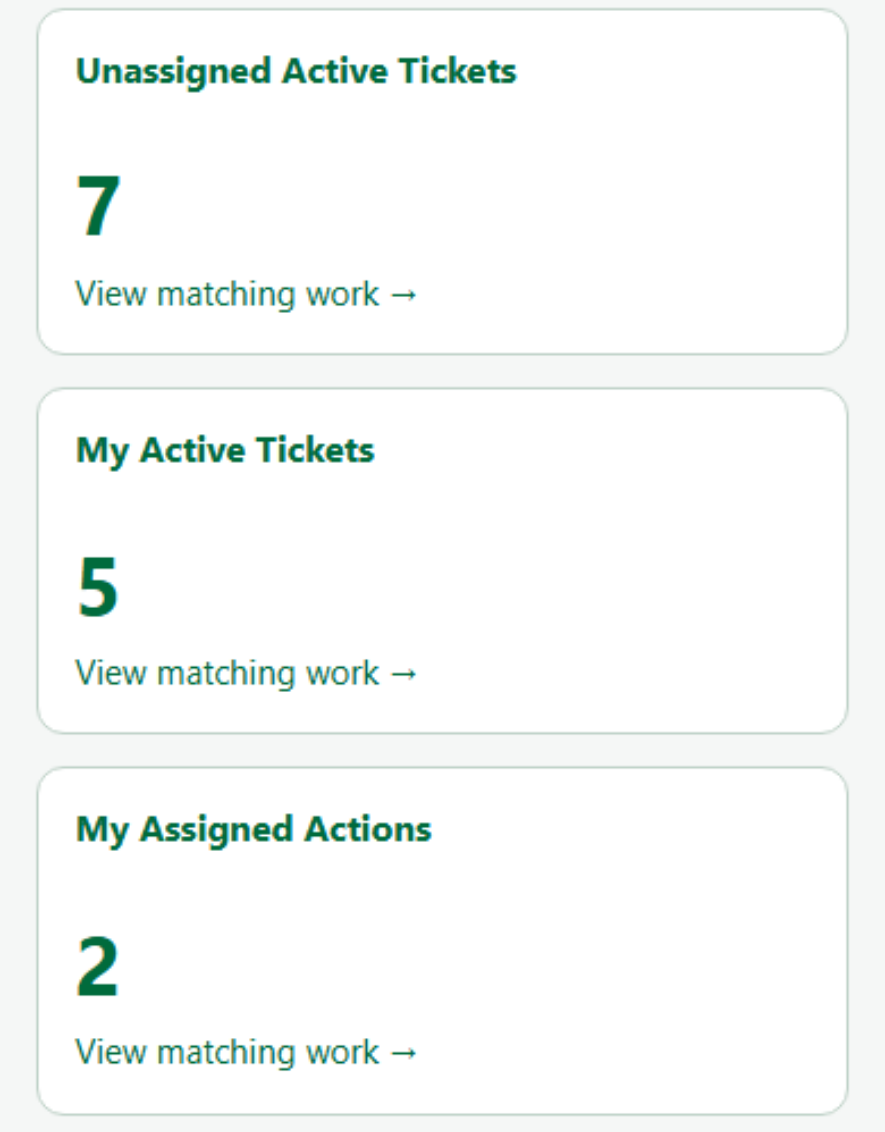
Operational snapshot - tablet

Figure 2 | staff-dashboard-768.png | application 0f169ea. Excerpt: (20, 180, 748, 1220). Original full PNG retained.



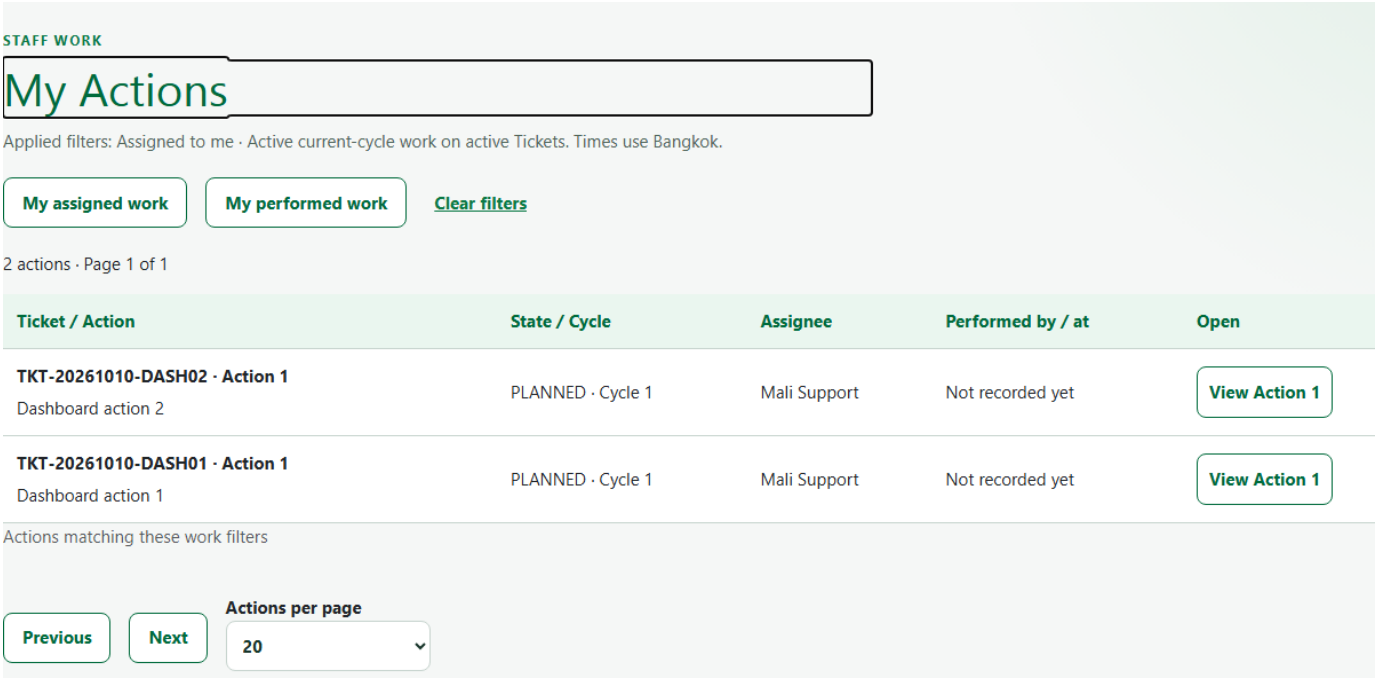
Operational snapshot - mobile work cards

Figure 3 | staff-dashboard-390.png | application 0f169ea. Excerpt: (0, 600, 390, 1100). Original full PNG retained.



Current-user assigned work and Ticket-local Action labels

Figure 4 | staff-work-1440.png | application 0f169ea. Excerpt: (120, 190, 1320, 780). Original full PNG retained.



Answer Part 6:

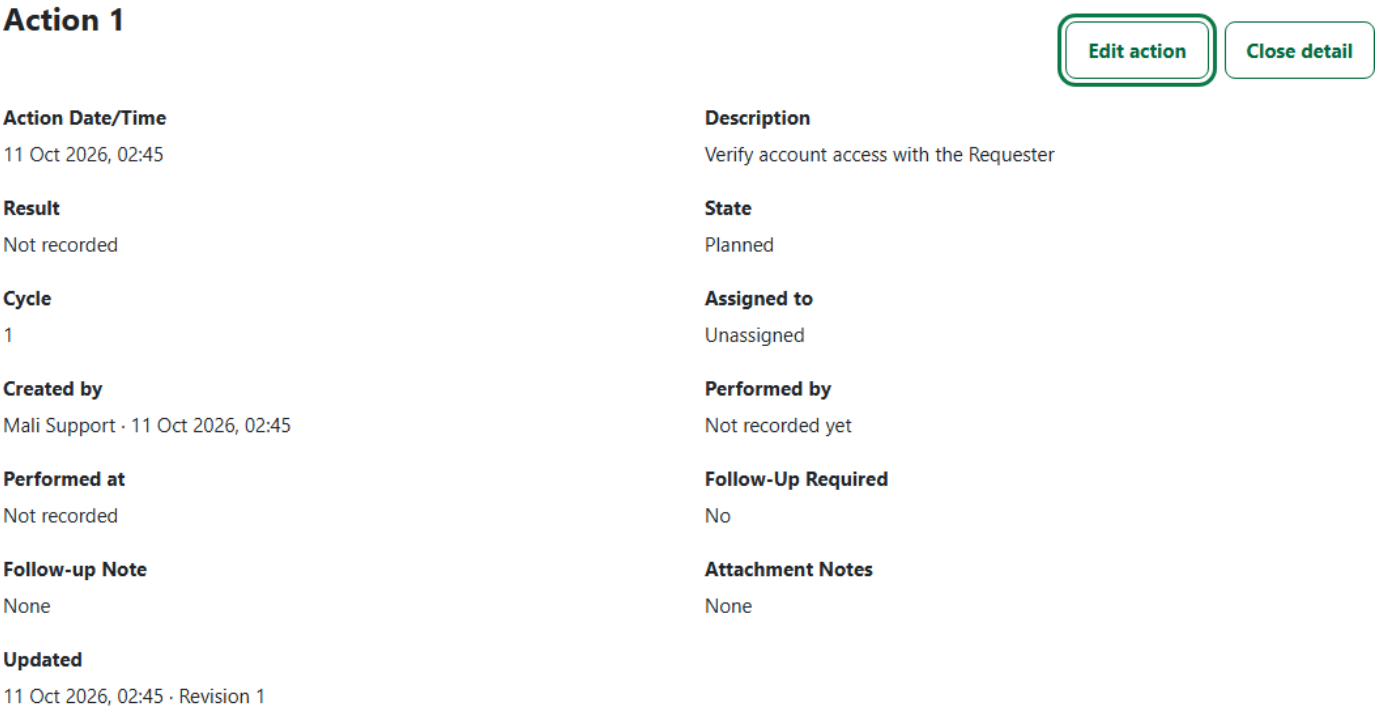
Working Actions Taken

Each Ticket has its own one-based Action display sequence; all creation-ordered records, cancelled work and prior cycles retain their positions. Global IDs still identify links, writes, receipts and immutable audit snapshots. Create starts Planned. Start changes only state to In progress; Complete requires a nonblank Result and explicit actual-performer confirmation. Cancel requires a reason. View/Edit, assignment and progress save independently.

The 49-case browser run exercises different actors and multiple actions, assignment, edit/start/complete/cancel, required result/note/reason, inactive-assignee and Requester denial, stale conflicts, safe/uncertain outcomes and exact replay without duplicate records/events. API tests directly enforce authorization; hiding controls is not the protection.

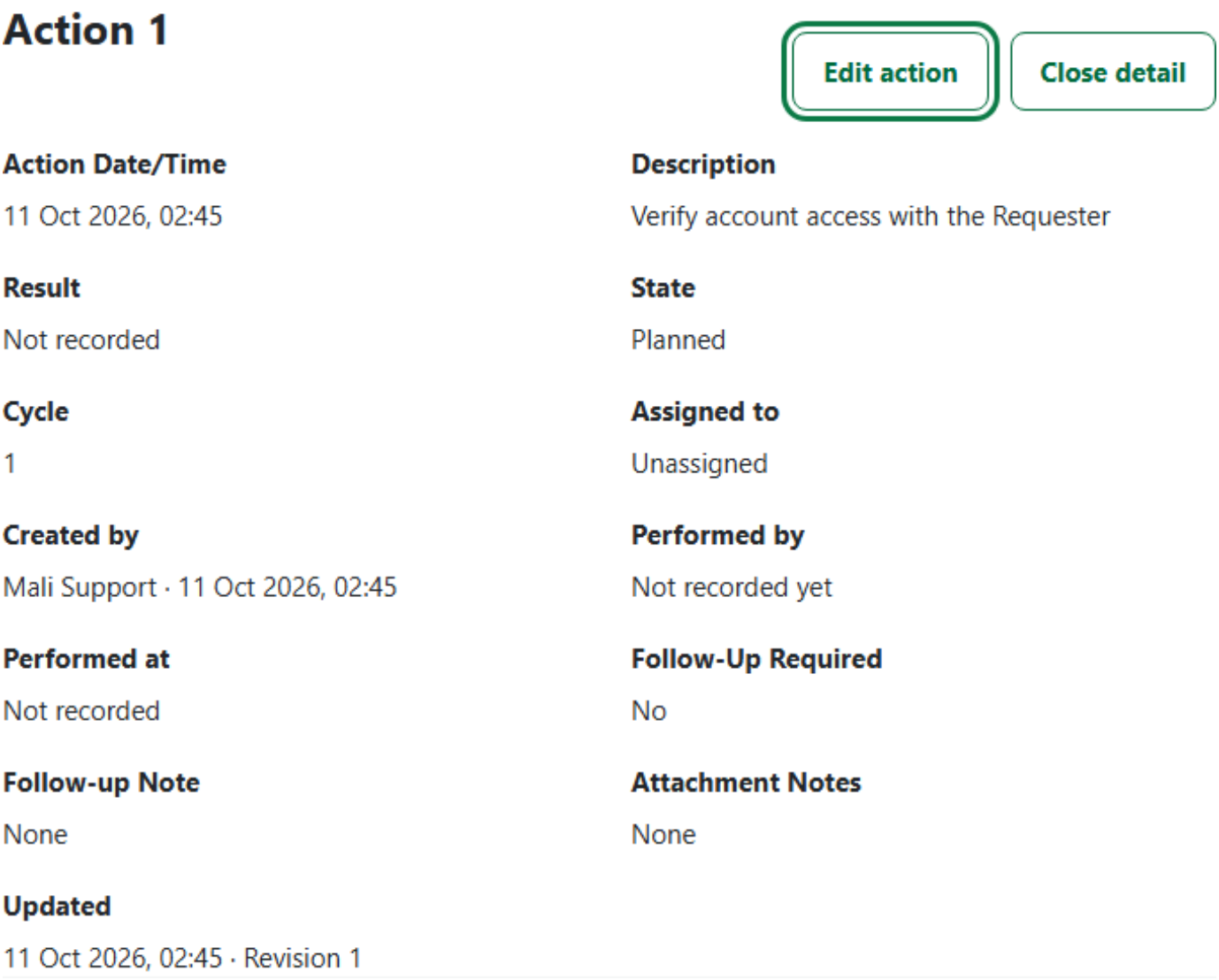
View-first action details - 1440px

Figure 5 | action-view-1440.png | application 0f169ea. Unmodified full region capture.



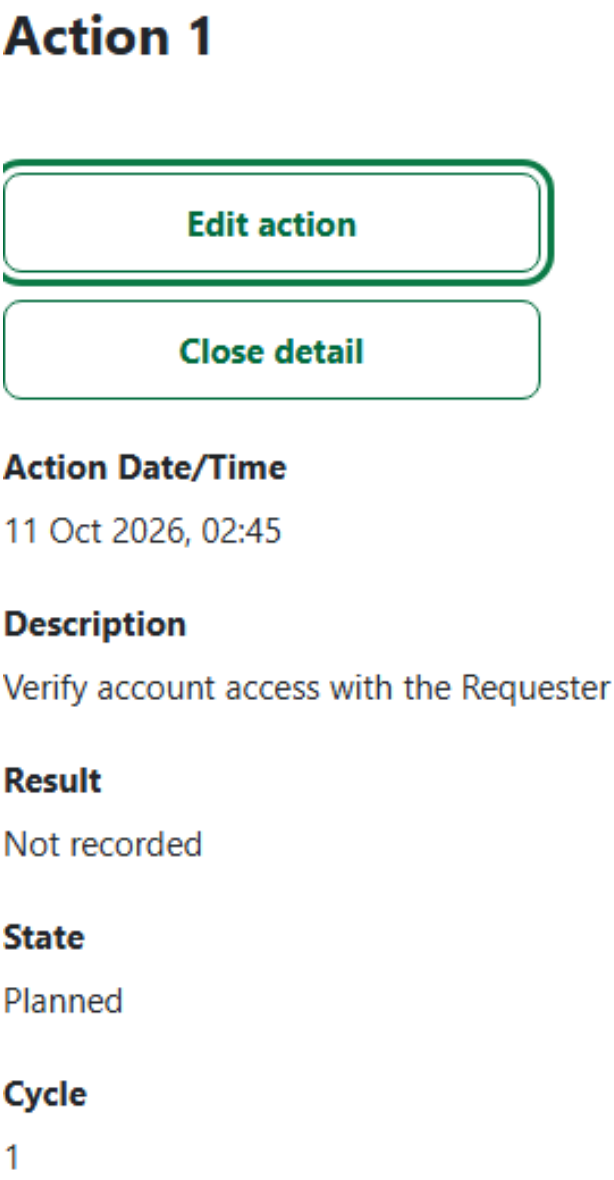
View-first action details - 768px

Figure 6 | action-view-768.png | application 0f169ea. Unmodified full region capture.



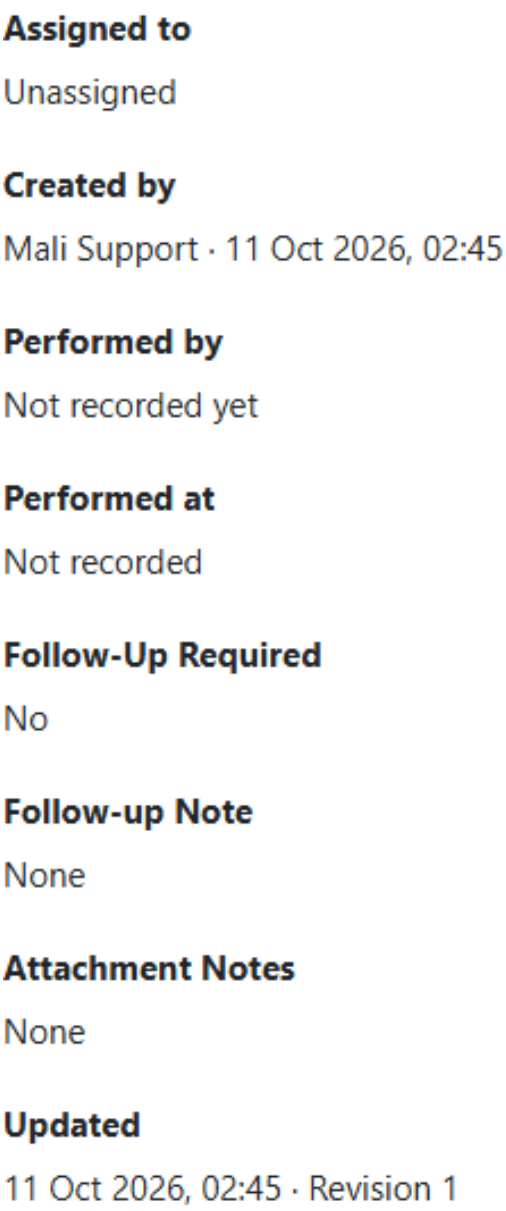
View-first action details - 390px - excerpt 1/2

Figure 7 | action-view-390.png | application 0f169ea. Excerpt: (0, 0, 320, 548). Original full PNG retained.



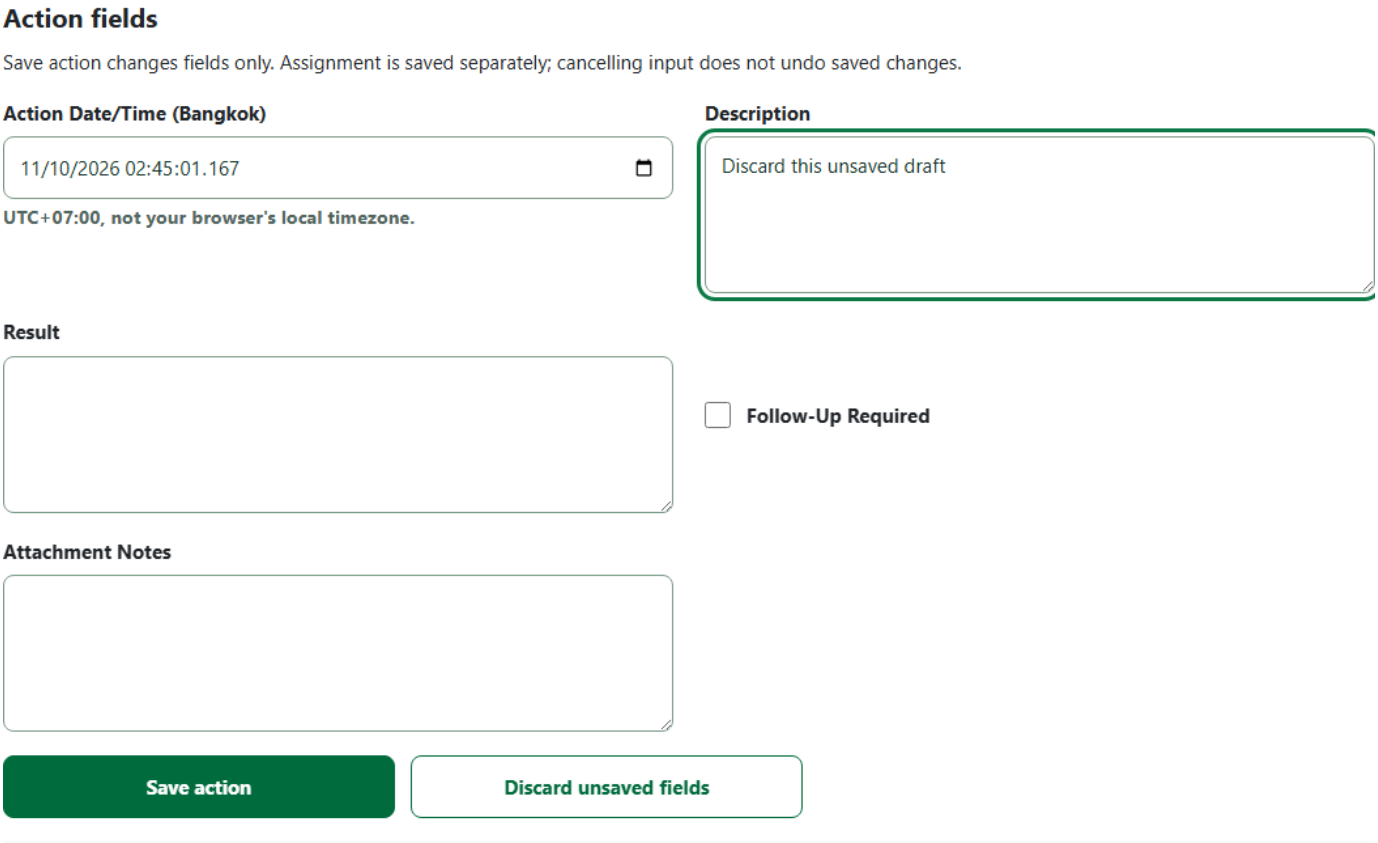
View-first action details - 390px - excerpt 2/2

Figure 8 | action-view-390.png | application 0f169ea. Excerpt: (0, 548, 320, 1096). Original full PNG retained.



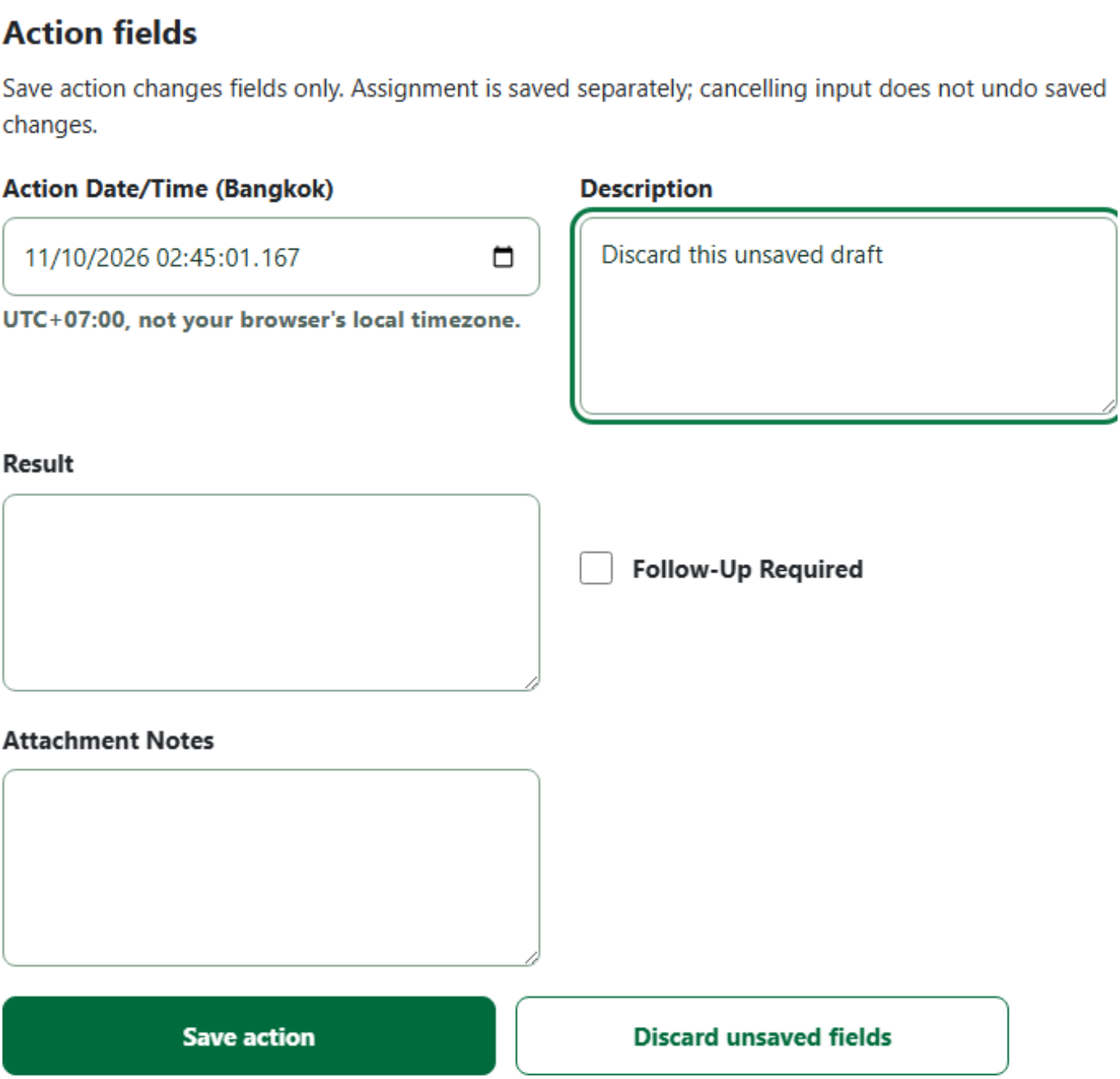
Explicit field editor and equal Save/Discard controls - 1440px

Figure 9 | action-edit-1440.png | application 0f169ea. Unmodified full region capture.



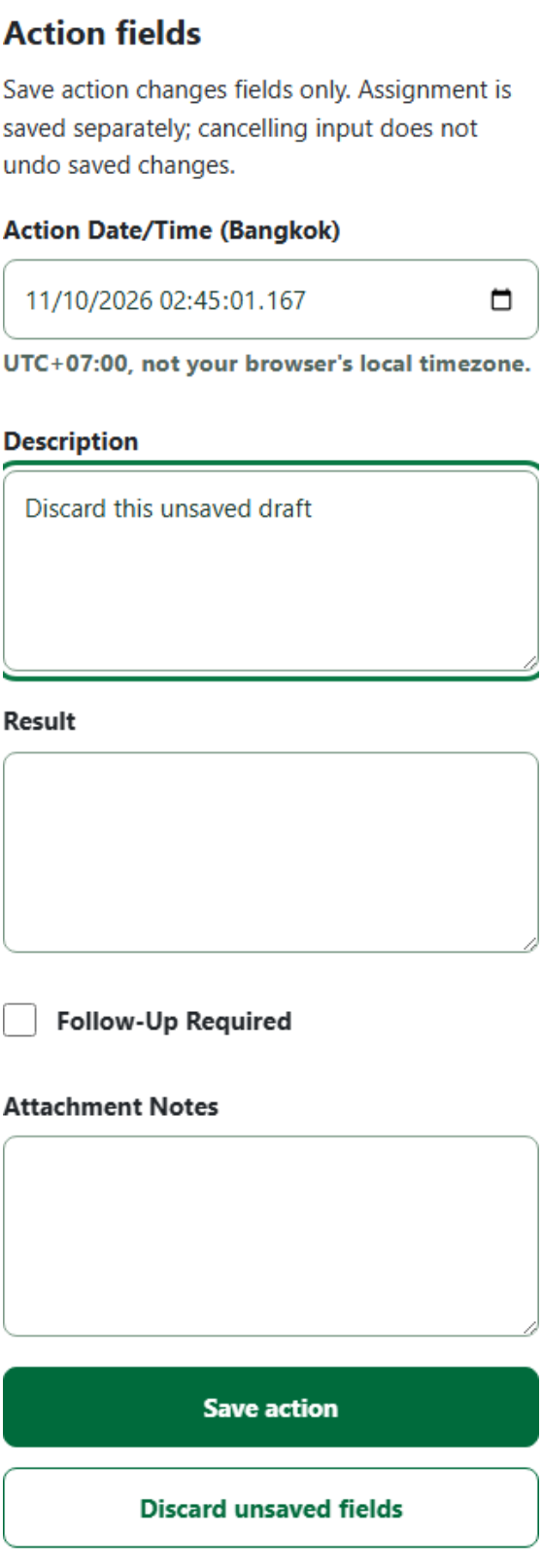
Explicit field editor and equal Save/Discard controls - 768px

Figure 10 | action-edit-768.png | application 0f169ea. Unmodified full region capture.



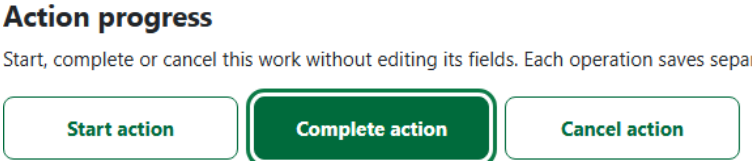
Explicit field editor and equal Save/Discard controls - 390px

Figure 11 | action-edit-390.png | application 0f169ea. Unmodified full region capture.



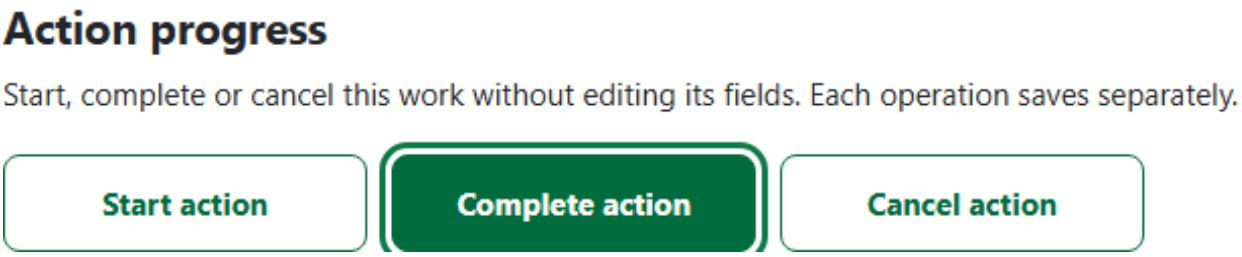
Progress independent of Edit - 1440px

Figure 12 | action-progress-1440.png | application 0f169ea. Unmodified full region capture.



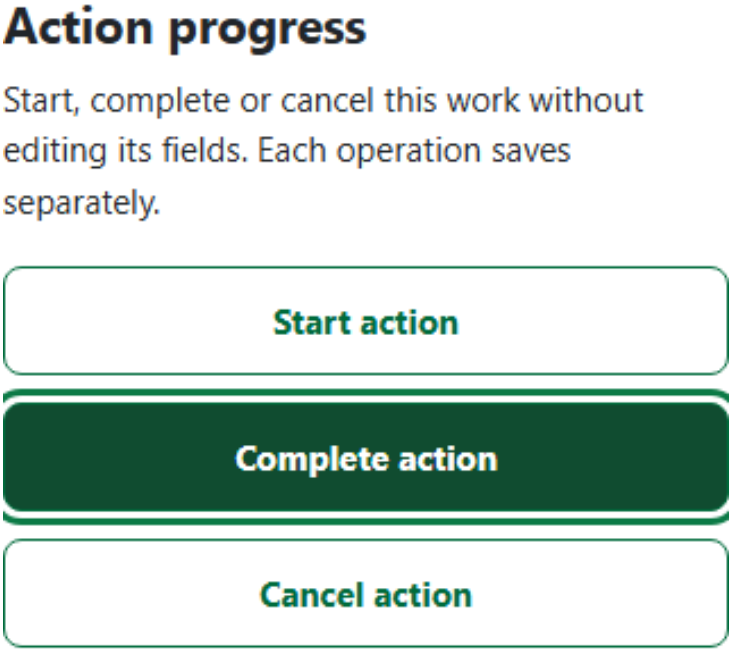
Progress independent of Edit - 768px

Figure 13 | action-progress-768.png | application 0f169ea. Unmodified full region capture.



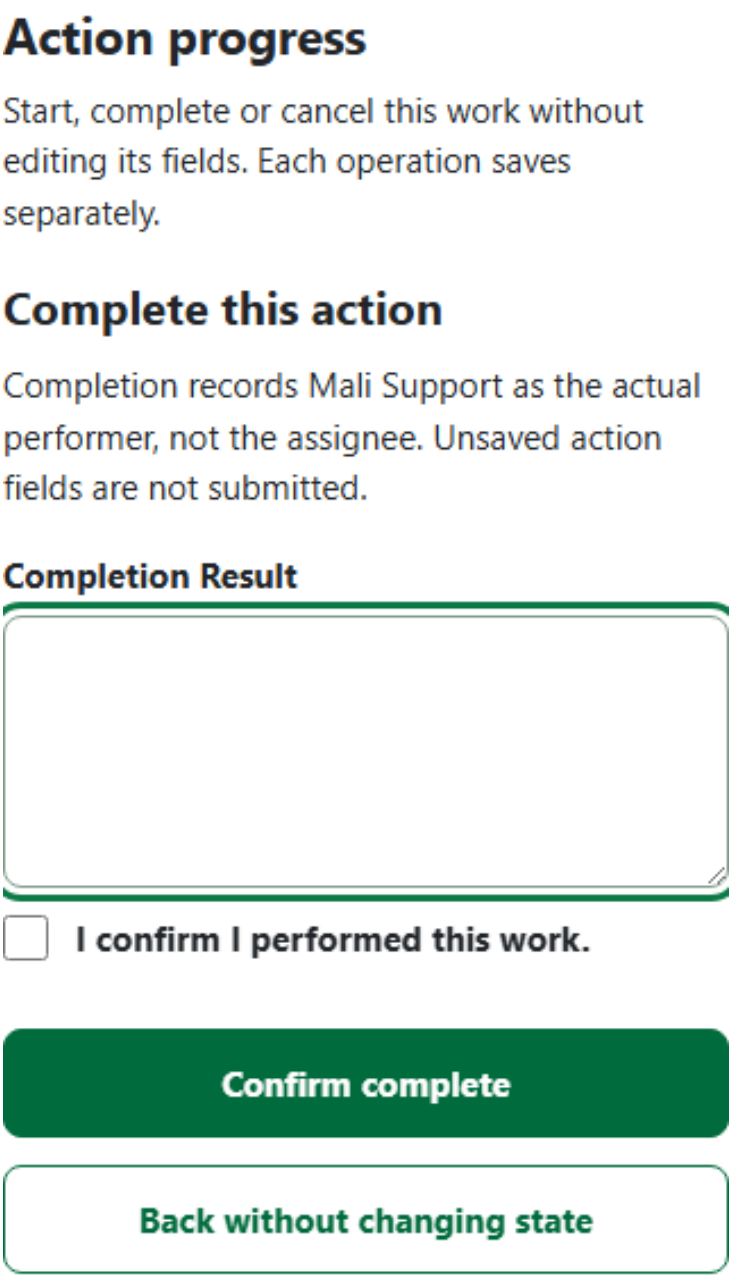
Progress independent of Edit - 390px

Figure 14 | action-progress-390.png | application 0f169ea. Unmodified full region capture.



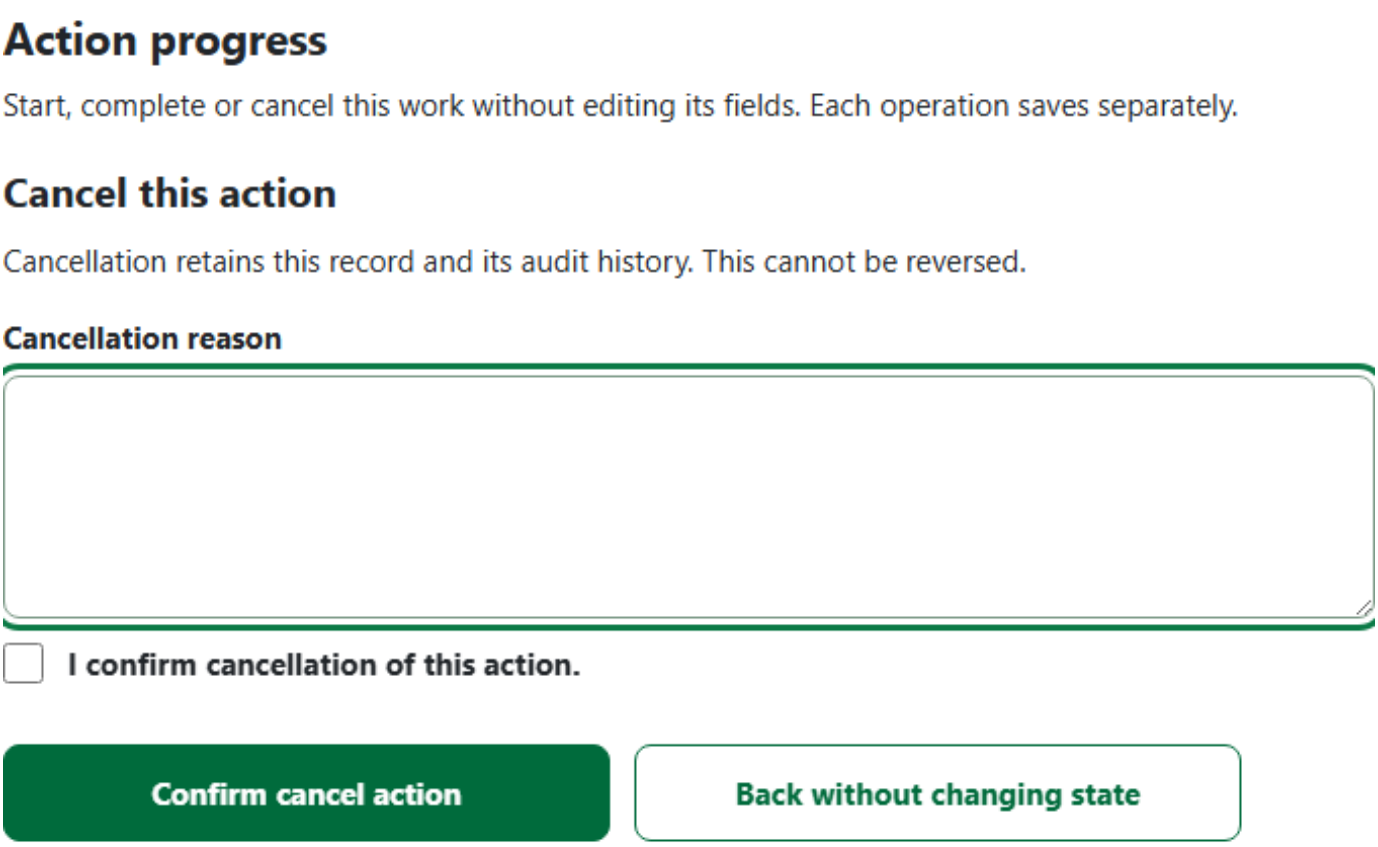
Completion Result and actual-performer confirmation

Figure 15 | action-confirm-complete-390.png | application 0f169ea. Unmodified full region capture.



Explicit cancellation reason and equal Back controls

Figure 16 | action-confirm-cancel-768.png | application 0f169ea. Unmodified full region capture.



Equal Create/Discard controls

Figure 17 | action-create-controls-1440.png | application 0f169ea. Unmodified full region capture.



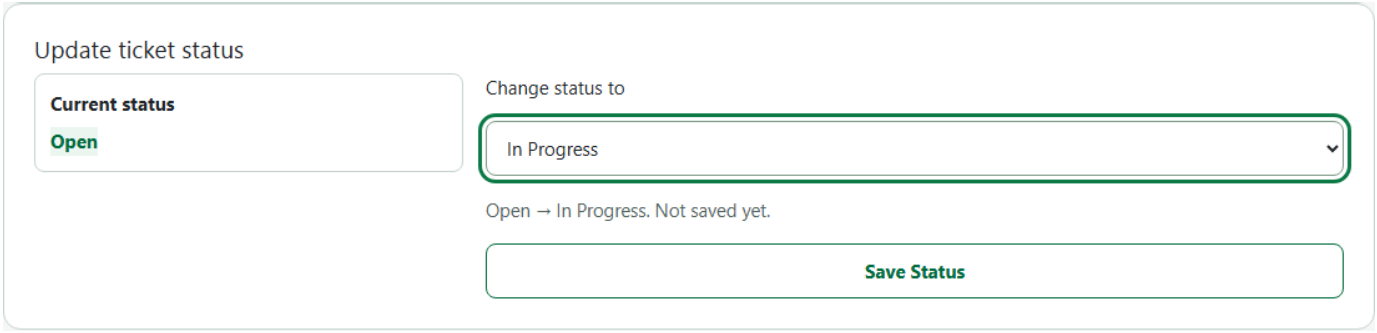
Answer Part 7:

Working Ticket workflow

Staff/Admin formal transitions follow the approved matrix; Requester indication is advisory. Whole-cycle resolution requires completed work, no active actions and no completed follow-up outstanding, including blockers beyond page one. Reopen advances the cycle and preserves previous work read-only. The API atomically rechecks versions/gates and appends history; lost responses block further writes until authoritative reload.

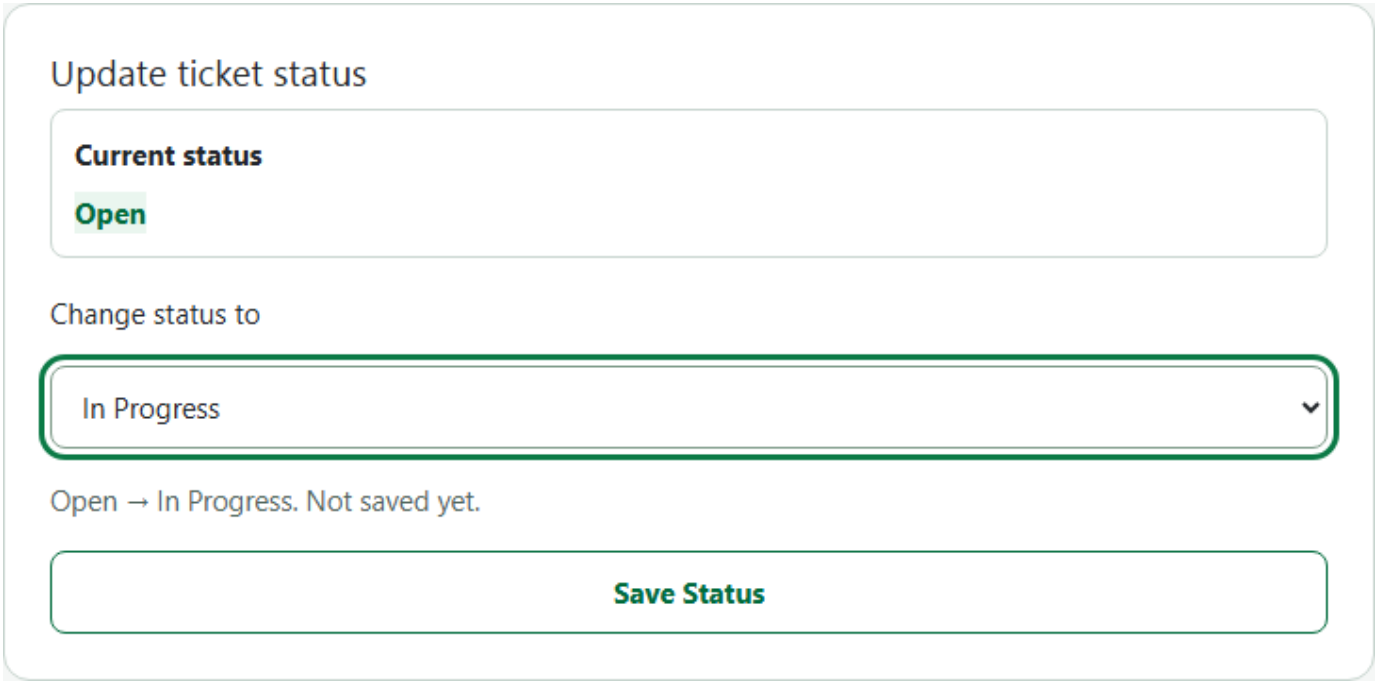
Saved status separate from proposed destination - 1440px

Figure 18 | status-selection-1440.png | application 0f169ea. Unmodified full region capture.



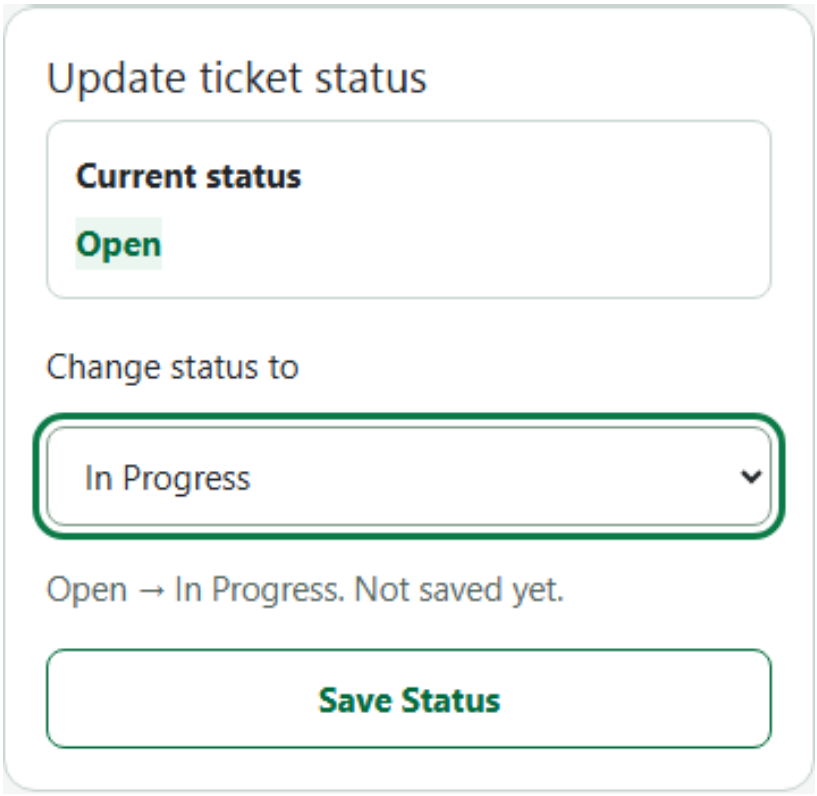
Saved status separate from proposed destination - 768px

Figure 19 | status-selection-768.png | application 0f169ea. Unmodified full region capture.



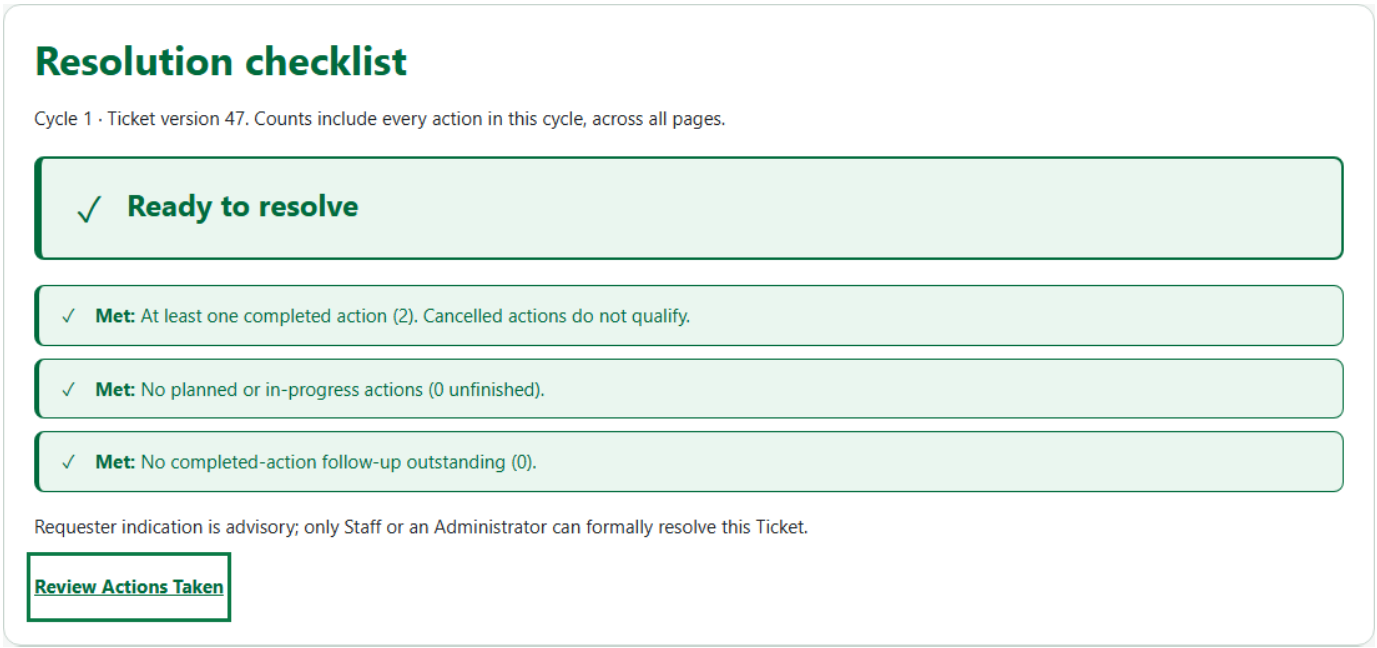
Saved status separate from proposed destination - 390px

Figure 20 | status-selection-390.png | application 0f169ea. Unmodified full region capture.



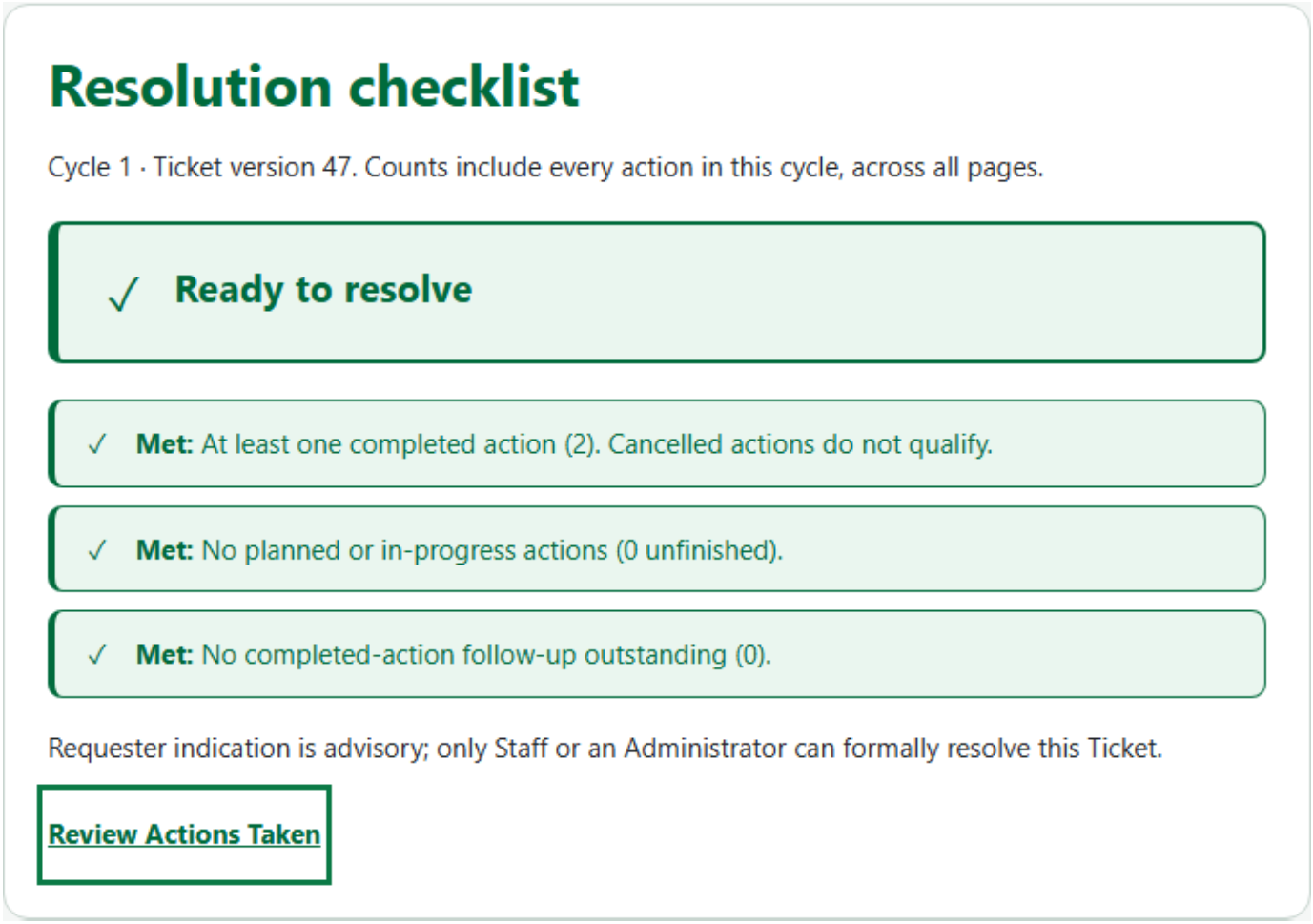
Whole-cycle readiness with text and icons - 1440px

Figure 21 | checklist-ready-1440.png | application 0f169ea. Unmodified full region capture.



Whole-cycle readiness with text and icons - 768px

Figure 22 | checklist-ready-768.png | application 0f169ea. Unmodified full region capture.



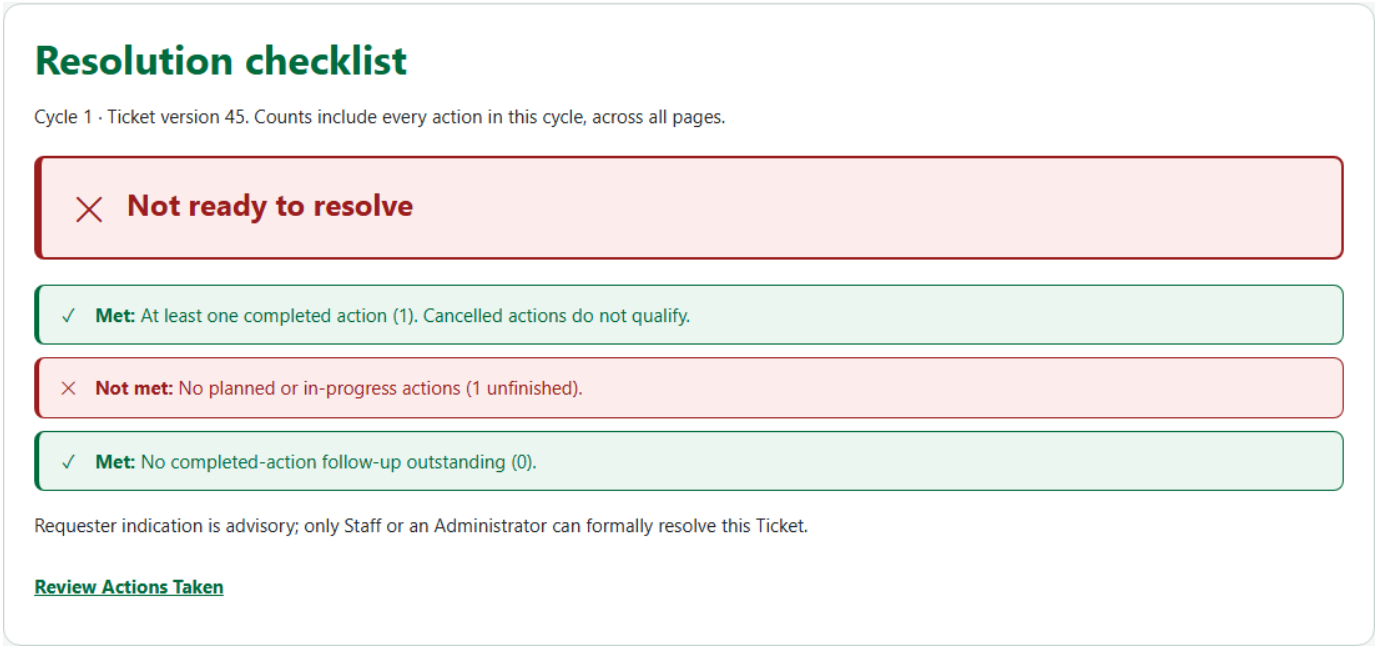
Whole-cycle readiness with text and icons - 390px

Figure 23 | checklist-ready-390.png | application 0f169ea. Unmodified full region capture.



Page-two unfinished work still blocks resolution

Figure 24 | later-page-blocker-1440.png | application 0f169ea. Excerpt: (0, 0, 1100, 515). Original full PNG retained.



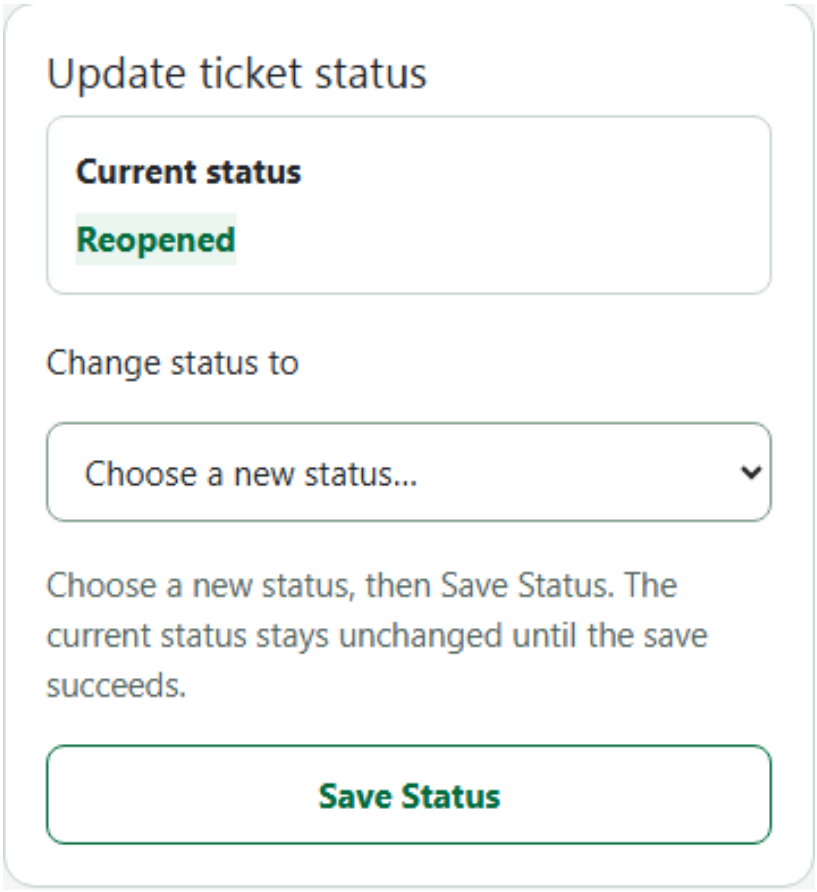
Reopened cycle starts with no completed work

Figure 25 | reopened-390.png | application 0f169ea. Excerpt: (0, 0, 358, 625). Original full PNG retained.



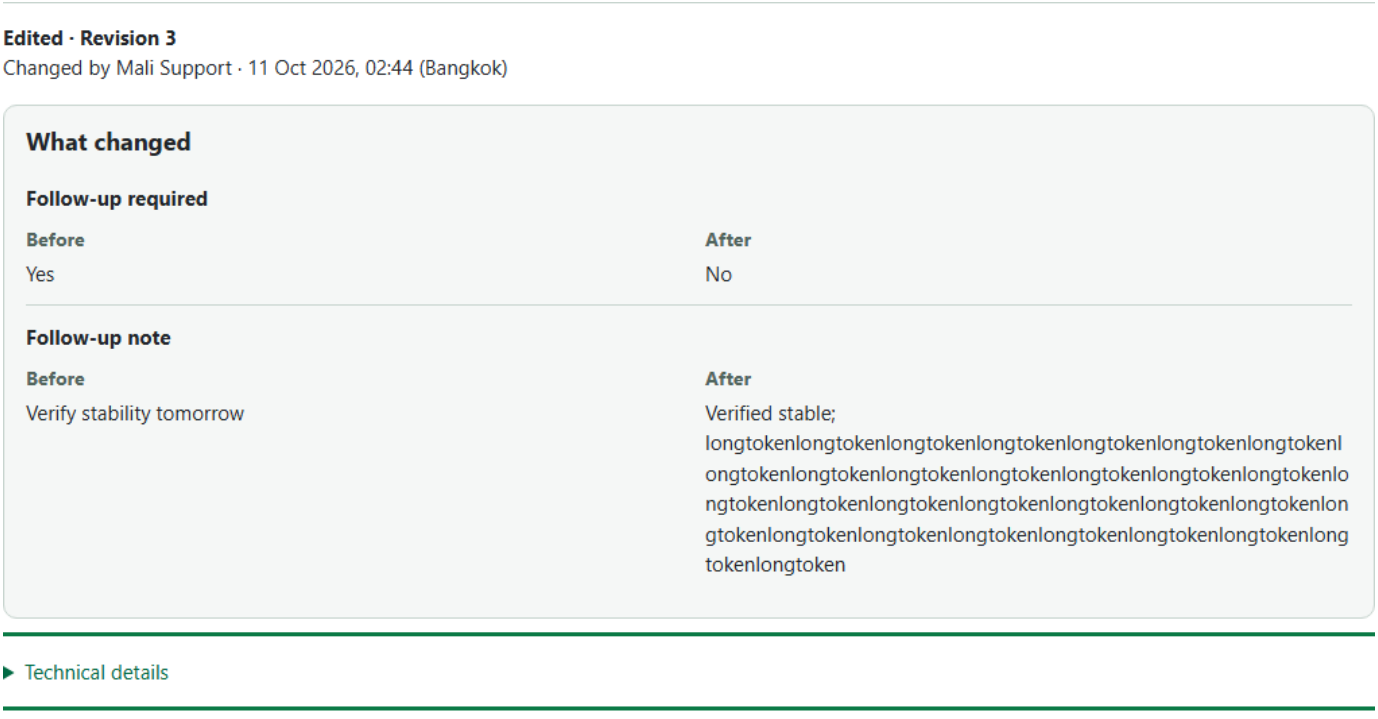
Saved Reopened status and retained transition choices

Figure 26 | reopened-390.png | application 0f169ea. Excerpt: (0, 1136, 358, 1526). Original full PNG retained.



Append-only changed values and technical disclosure - 1440px

Figure 27 | audit-readable-staff-1440.png | application 0f169ea. Unmodified full region capture.


Edited · Revision 3
Changed by Mali Support · 11 Oct 2026, 02:44 (Bangkok)
What changed
Follow-up required
Before
Yes
After
No
Follow-up note
Before
Verify stability tomorrow
After
Verified stable; longtokenlongtokenlongtokenlong

Append-only changed values and technical disclosure - 768px

Figure 28 | audit-readable-staff-768.png | application 0f169ea. Unmodified full region capture.

Edited · Revision 3

Changed by Mali Support · 11 Oct 2026, 02:44 (Bangkok)

What changed

Follow-up required

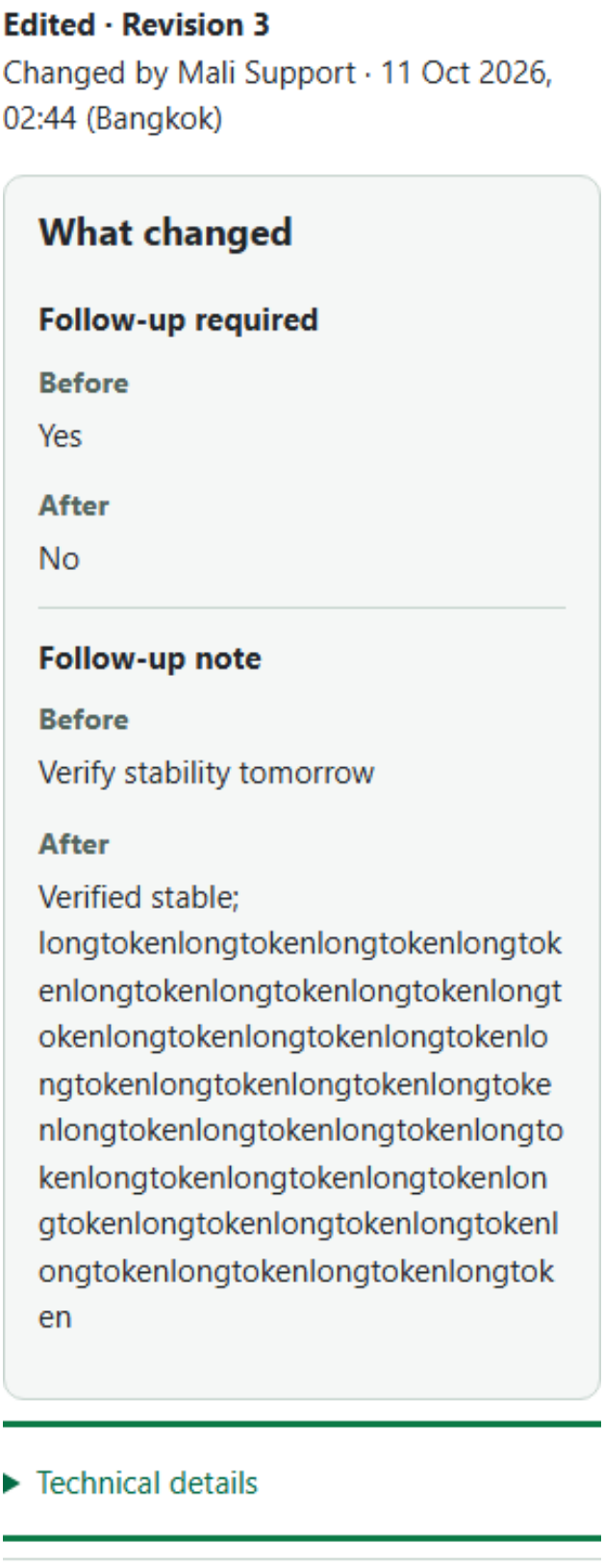
Before	After
Yes	No

Follow-up note

Before	After
Verify stability tomorrow	Verified stable; longtokenlongtokenlongtokenlong

Append-only changed values and technical disclosure - 390px

Figure 29 | audit-readable-staff-390.png | application 0f169ea. Unmodified full region capture.


Edited · Revision 3
Changed by Mali Support · 11 Oct 2026, 02:44 (Bangkok)
What changed
Follow-up required
Before
Yes
After
No
Follow-up note
Before
Verify stability tomorrow
After
Verified stable; longtokenlongtokenlongtokenlong

Answer Part 8:

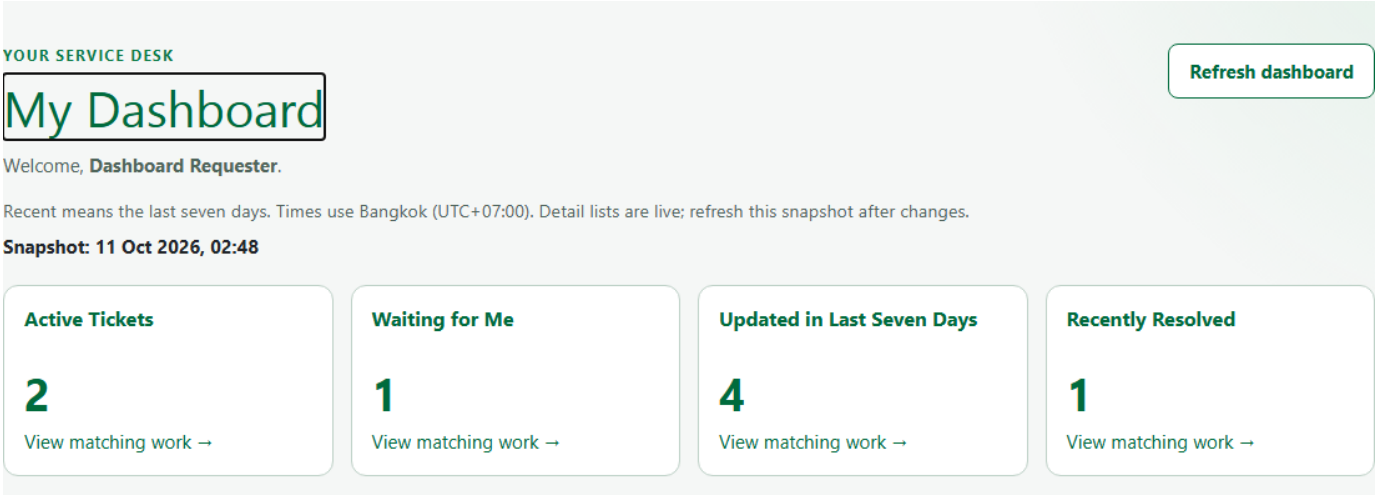
Requester dashboard and final regression

Requester metrics always include the authenticated owner. Active means New/Open/In Progress/Waiting for Requester/Reopened. Recent is the inclusive elapsed seven-day UTC interval, displayed in Bangkok. Legacy unknown resolution dates are not invented. Empty accounts receive real zeroes; wrong-role/ownership requests are denied. Recent/attention links reach owned filtered lists.

Requester fixture / metric	API	Independent rows	Live list
Populated / activeTickets	2	2	2
Populated / waitingForMe	1	1	1
Populated / recentlyUpdated	4	4	4
Populated / recentlyResolved	1	1	1
Empty / activeTickets	0	0	0
Empty / waitingForMe	0	0	0
Empty / recentlyUpdated	0	0	0
Empty / recentlyResolved	0	0	0

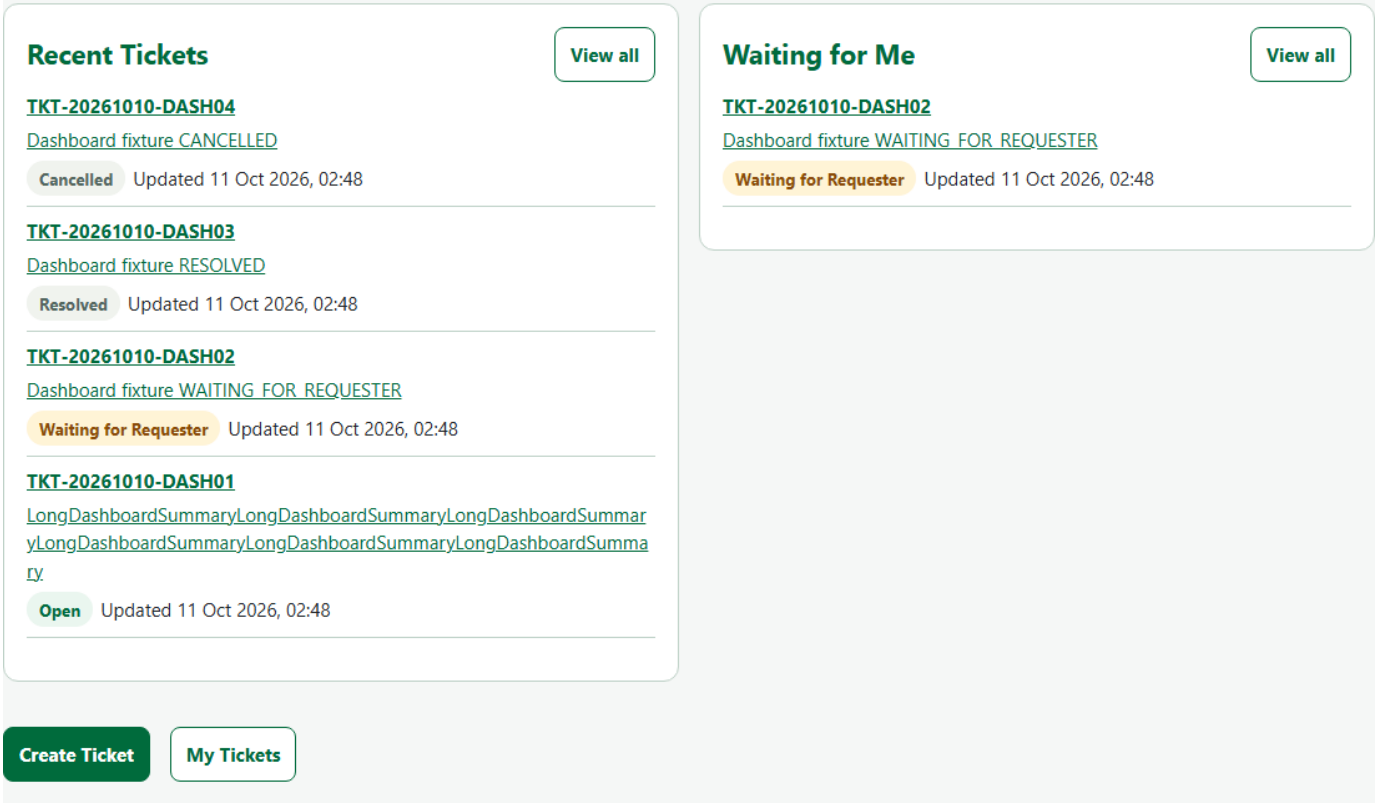
Owned dashboard - desktop metrics

Figure 30 | requester-dashboard-1440.png | application 0f169ea. Excerpt: (170, 180, 1270, 575). Original full PNG retained.



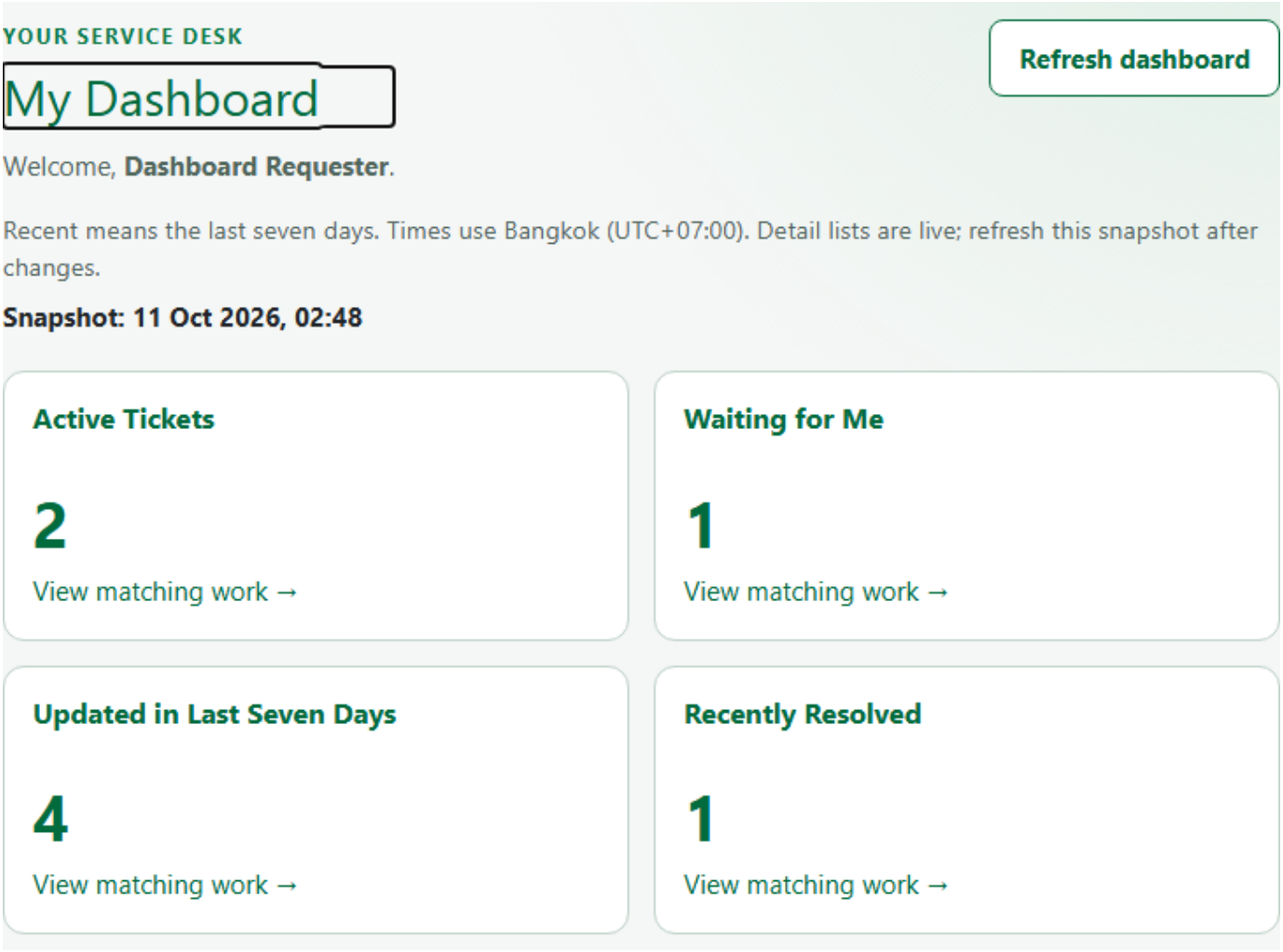
Owned dashboard - desktop recent/attention lists

Figure 31 | requester-dashboard-1440.png | application 0f169ea. Excerpt: (170, 575, 1270, 1220). Original full PNG retained.



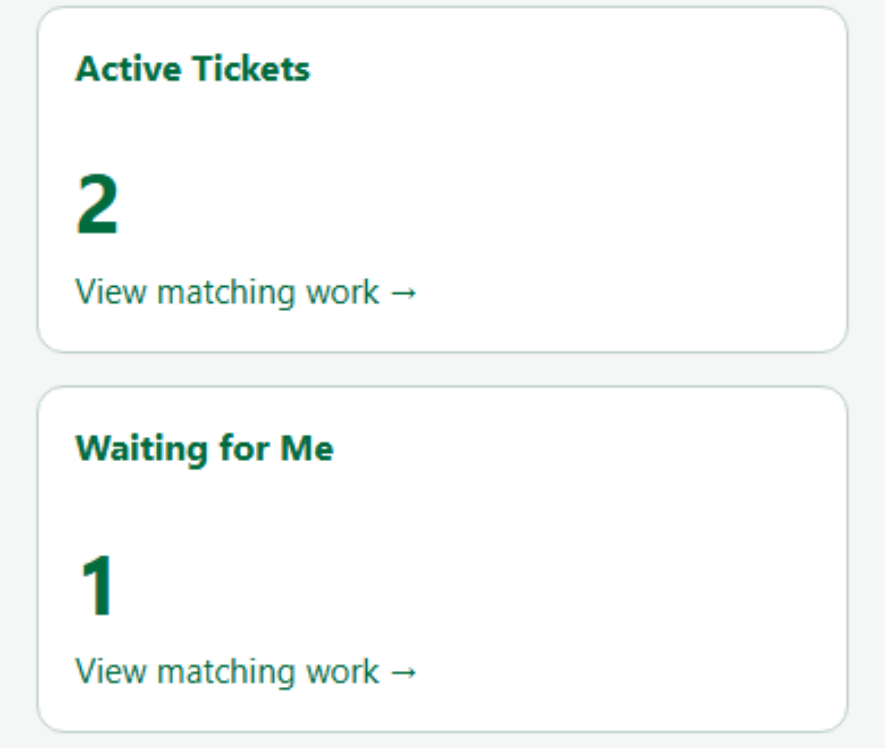
Owned dashboard - tablet metrics

Figure 32 | requester-dashboard-768.png | application 0f169ea. Excerpt: (20, 180, 748, 720). Original full PNG retained.



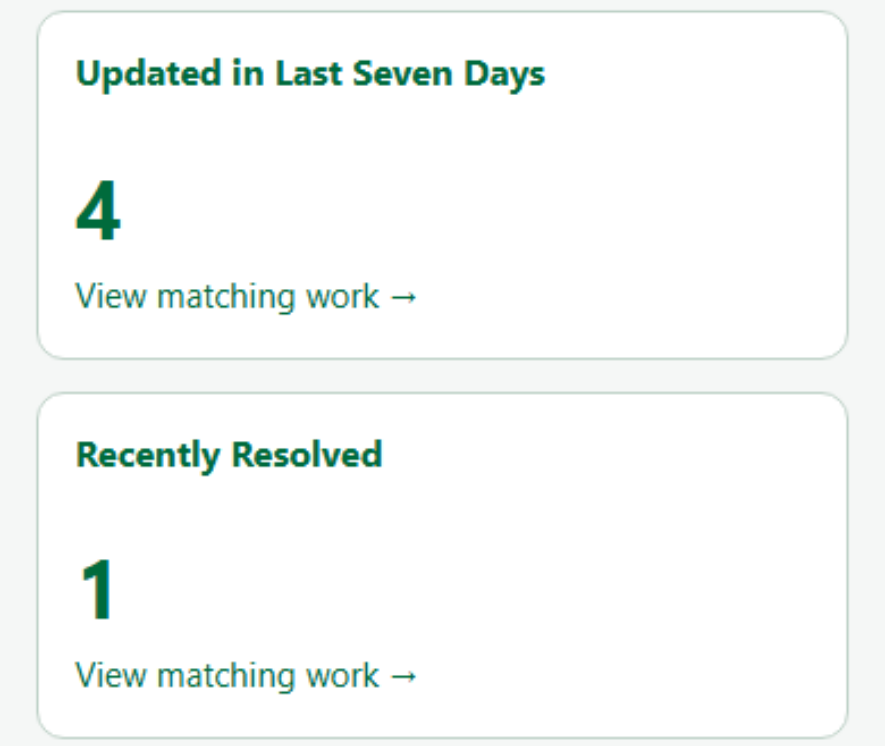
Owned dashboard - mobile, first work cards

Figure 33 | requester-dashboard-390.png | application 0f169ea. Excerpt: (0, 545, 390, 875). Original full PNG retained.



Owned dashboard - mobile, recent work cards

Figure 34 | requester-dashboard-390.png | application 0f169ea. Excerpt: (0, 877, 390, 1206). Original full PNG retained.

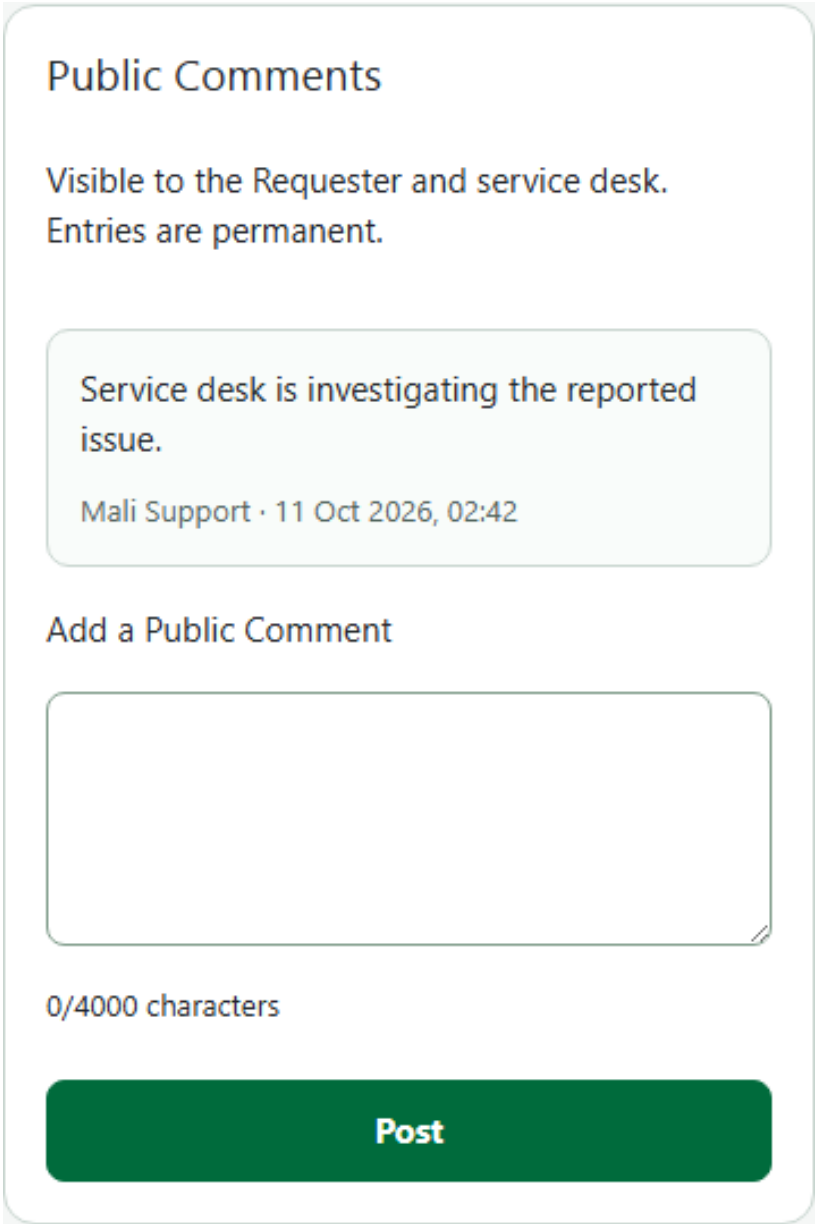


Representative preserved earlier behavior

Area	Actual final-main checks
Authentication / accounts	Login/change/logout/revocation and Admin lifecycle/final-admin/stale safeguards
Requester Ticket / attachments	Create/find/open; upload/download/remove; failed initial upload cannot falsely claim rollback or duplicate creation
Comments / Notes / Staff	Append-only authored Public Comments; Staff operations; Requester Notes route returns 403 without private text
Actions / transitions	Owned read-only shared history, immutable audit, terminal/prior-cycle locks; exact record deep links

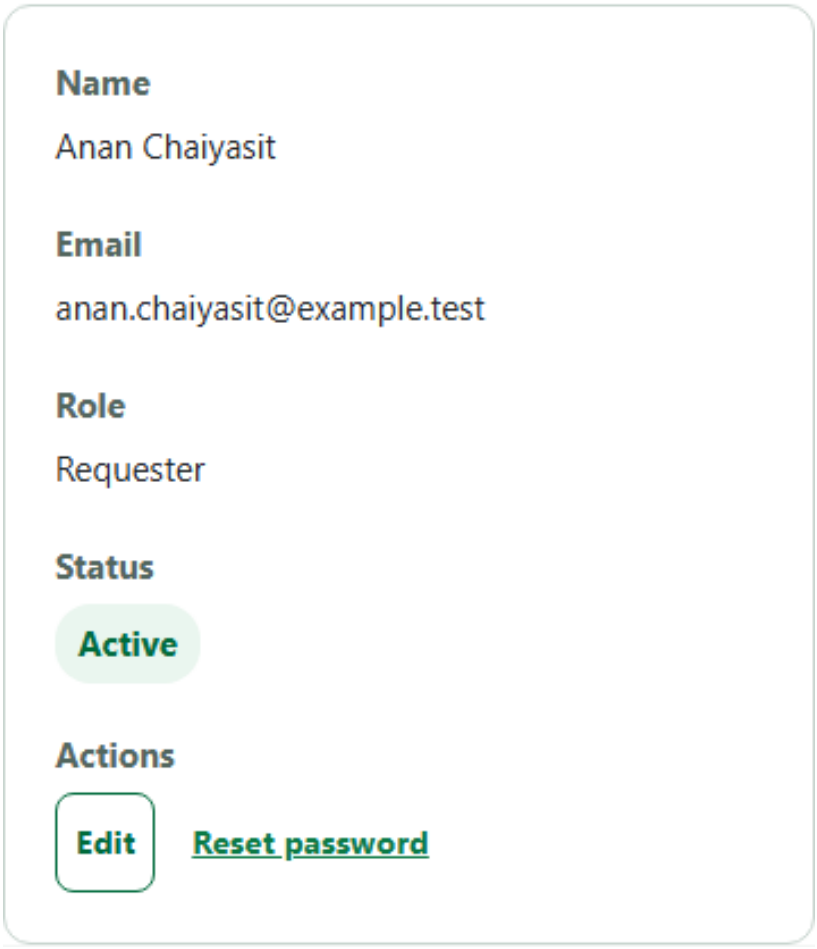
Shared comment remains readable on mobile

Figure 35 | public-comments-mobile.png | application 0f169ea. Unmodified full region capture.



Administrator account card - mobile

Figure 36 | user-row-mobile.png | application 0f169ea. Unmodified full region capture.



Answer Part 9:

Zen Green, responsive and accessibility verification

Rendered ui-spec.md - selected current sections

Published reviewed source: ui-spec.md. The complete maintained Markdown is embedded, including the later author-approved closeout. Published-version links and current embedded closeout text are explicitly distinguished; dated history is retained in the attachment.

Lab 4 UI Specification

Status: Approved contract and peer-accepted foundations/Action UI in PRs #45-47. Issue #42 adds the current-cycle checklist, allowed status destinations, cancellation guidance, reopen labels and paged read-only transition history. Counts come from the whole-cycle ActionPage summary and must match the parent's version/cycle; loading, mixed snapshots and busy/unknown saves never mean ready. Lost status responses freeze writes until an explicit authoritative reload. Issue #42 screens were peer-approved in PR #48 and merged as 00fddc1. Dashboard screens (#43) were author-accepted and peer-approved/merged in PR #49 (4660ac6). Clean main 0f169ea passed the final gate and supplementary screen/query checks on October 11; fresh evidence is in artifacts/lab-04/final-main/. PR #52 published the reviewed evidence as main 72314d5. The author subsequently approved the report and authorized final assembly. Automated/agent visual checks, author report approval and a new manual demo are distinct; the latter is not claimed. Business rules: specification.md; requests: api-spec.md.

Navigation and routes

Implemented #43: Requester home/brand use /dashboard; Staff/Admin use /staff/dashboard. Existing list/create/queue/account navigation is preserved, with aria-current on the active route. Counts are labelled links with large numbers; status/priority links include explicit zeros. Pending initial reads show loading, not placeholder zeros. Failed refreshes label the retained successful snapshot Not refreshed; forbidden responses clear protected metrics. User-keyed views ignore late old-identity responses. Requester cards use four/two/one columns across desktop/tablet/mobile; Staff cards use three/two/one, with aligned metric numbers. Count links and quick actions remain at least 44px tall. Captured time pairs, status-group/owner/assignment filters and paging round-trip through URLs on reload/Back. Invalid URLs show safe feedback and no unfiltered results. My Actions links open the exact Action record through tab/actionId.

Route	Role	Purpose
/dashboard	Requester	Owned dashboard and role home
/staff/dashboard	Staff/Admin	Operational dashboard and role home
/staff/actions	Staff/Admin	Paged work list reached from dashboard
/tickets, /tickets/new, /tickets/:id	Requester	Existing flows; detail adds read-only Actions/history
/staff/tickets, /staff/tickets/:id	Staff/Admin	Existing queue/detail; detail adds Actions/history
/admin/users	Admin	Existing account management
/login, /change-password	Existing allowed roles	Preserved authentication/account behavior

Requester navigation: Dashboard, My Tickets, Create Ticket. Staff: Dashboard, Ticket Queue. Admin adds Users. Preserve Change password above current identity and Sign out. Brand link returns to role dashboard. aria-current marks the active destination; mobile navigation retains keyboard operation and visible focus. Initial-password accounts remain gated before any dashboard. A wrong-role URL uses Access denied; expired session returns to login. Preserve query strings.

Responsive and accessibility rules

Reuse green tokens and text-bearing status/priority badges. Approved compact workspace body 15px/helper >=14px, controls with >=44px touch targets, visible focus and semantic headings, table captions, labels, errors and buttons. Dashboard cards wrap; no color-only cues. At 1440px use compact card grids/tables; at 768px reduce columns/use readable cards; at 390px use one column, full-width fields and safely wrapping long text/identities. No page horizontal overflow, clipping or overlapping controls. Read-only text remains readable, not merely disabled low-contrast inputs.

Completed recorded verification - separate from final author acceptance

Final-revision visual and accessibility verification

Revision 0f169ea; full gate plus separate guarded healthy-screen audit, October 11. Original captures: 192 indexed PNGs. Environment 5183/3100; fixture-only identities. This records automated verification and agent visual review. The author separately approved the report and authorized finalization on October 11; no new personal demo or formal manual screen-reader certification is claimed.

Check	Actual evidence / result	Verification level
Consistent green hierarchy, compact dashboards and emphasized important headings	Staff/Requester 1440/768/390; Action/view/edit/progress and checklist close-ups visually reviewed	Agent visual + computed browser styles
Dashboard values and exact drill-downs	36 raw-row/API/list comparisons; bounded summary order; captured time pairs	Independent query + browser
Three-width major screens	Dashboards, My Actions, actions/read-only history, opening/status/checklist/audit; legacy login/create/list/detail/queue/users captures retained	Browser overflow/layout assertions + representative visual QA
Fields versus view-only information	View-first, explicit Edit/Close, readable prior-cycle/Requester records, independent assignment/state/field saves	Component/API/E2E + inspected view/editor panels
Non-color status cues and resolution feedback	Text/check/cross plus green/red checklist; cycle/version and saved status versus unsaved target	Browser/component + visual QA
Required result/note/reason and immutable actual performer	Complete/cancel confirmation, conditional notes/corrections; inactive assignee/Requester denial	API/component/E2E; representative confirmation/audit captures
Keyboard and focus	New/Edit/Close/Back focus handoff, inline confirmation, dashboard links, deep-link target, pagination	Existing real-browser/component assertions
Equal/aligned controls	Create/edit/Discard, progress, complete/cancel confirmation at all three widths; 44/48px targets	Real-browser dimensions and representative visual QA
Long text, clipping, overlap, overflow	Long unbroken summary/audit note wraps; full suite root/body width assertions; 24 extra healthy-screen width checks	Browser + representative image review
Legacy regression/privacy	Auth/password/accounts, creation/uploads/files, Public Comments, private Notes, queue/detail remain covered	Complete 298/211/49 run; inspected legacy panels
Healthy runtime errors and navigation	8 screens x 3 widths: zero unexpected console/page errors; auth-only signed-out 401 probes explicitly retained	Supplementary browser instrumentation
Recovery, empty, forbidden, stale and uncertain feedback	Empty/refresh-denied dashboard, failed upload, exact retry, conflict reload, page-two gate and history retry	Existing API/component/E2E assertions and retained captures
Author report acceptance	Author asked whether the reviewed report was sufficient, then authorized finalization	Report approval; no new personal demo or formal accessibility audit claimed

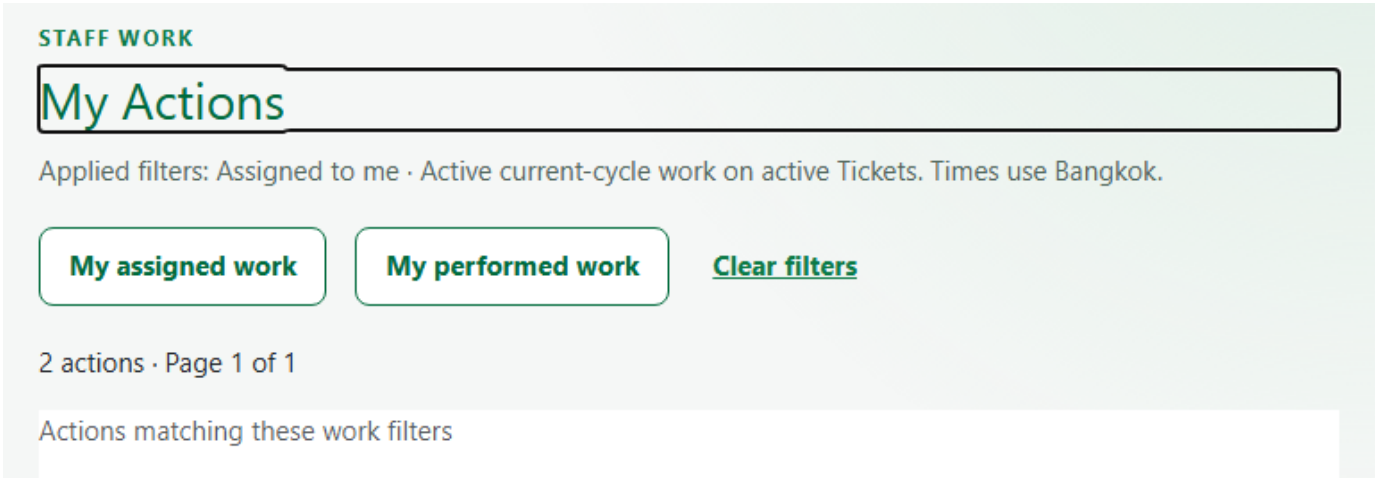
Check	Actual evidence / result	Verification level
Manual screen-reader / full WCAG certification	Not performed; no certification claimed	Out of this recorded verification

PDF close-ups are identified as excerpts of original images rather than silently shrinking very long mobile pages. Source paths, hashes and capture times are in captured-screenshots.json. Healthy supplementary captures and mutable full-suite captures belong to different isolated fixture runs; their counts must not be mixed.

The fresh 192-image index covers three-width major Lab 4 screens and earlier role screens. Original assets have hashes and timestamps. Figures are readable full regions or labelled excerpts, not miniature full-page thumbnails. Application behavior/style is unchanged from the reviewed release. Author report approval is recorded in the completion attachment. No new manual demo or formal accessibility certification is claimed.

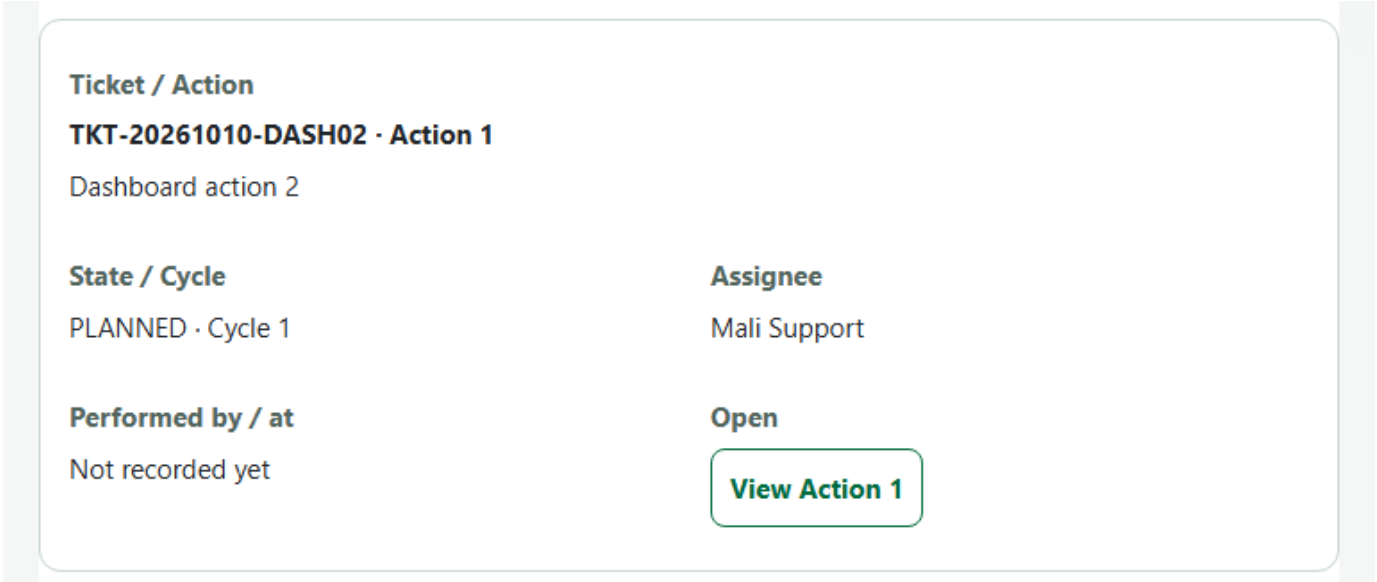
Responsive My Actions - tablet navigation

Figure 37 | staff-work-768.png | application 0f169ea. Excerpt: (0, 180, 768, 445). Original full PNG retained.



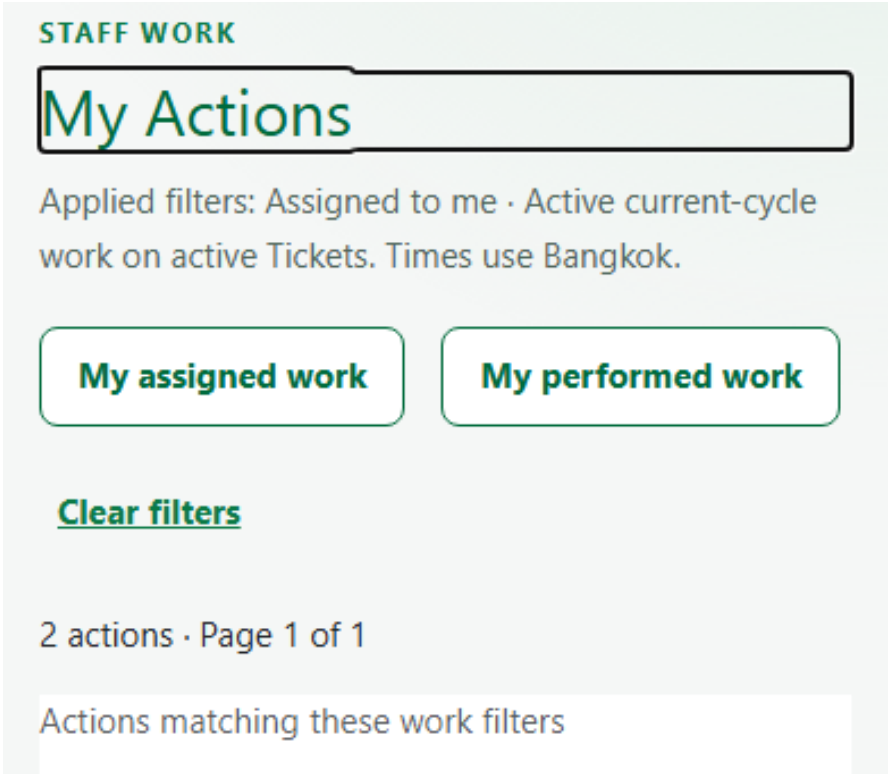
Responsive My Actions - tablet work card

Figure 38 | staff-work-768.png | application 0f169ea. Excerpt: (0, 452, 768, 780). Original full PNG retained.



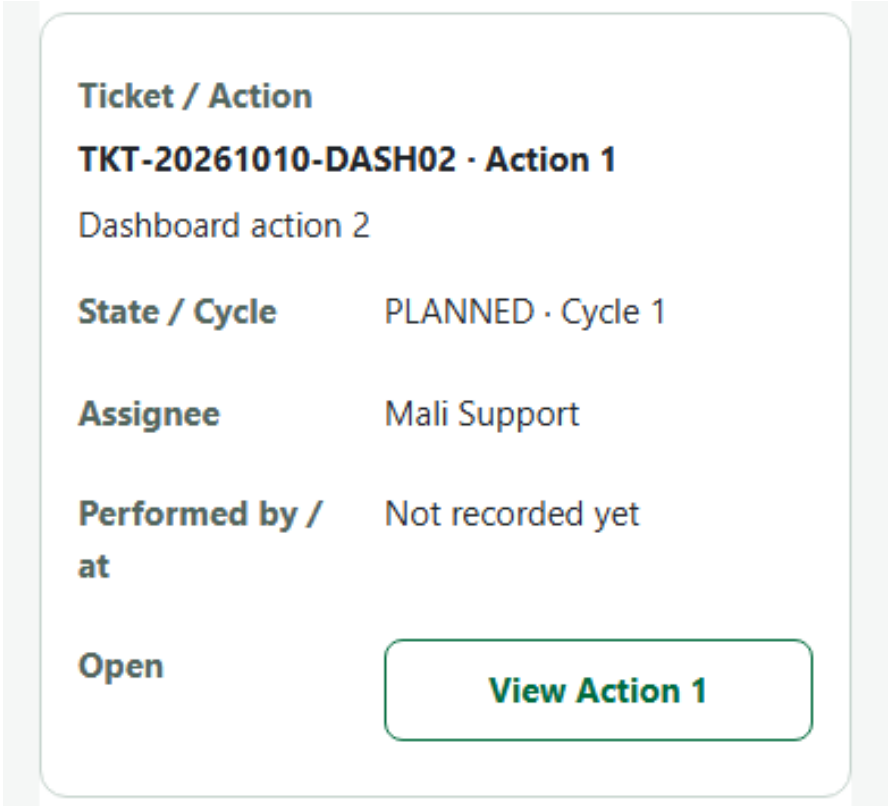
Responsive My Actions - mobile navigation

Figure 39 | staff-work-390.png | application 0f169ea. Excerpt: (0, 310, 390, 650). Original full PNG retained.



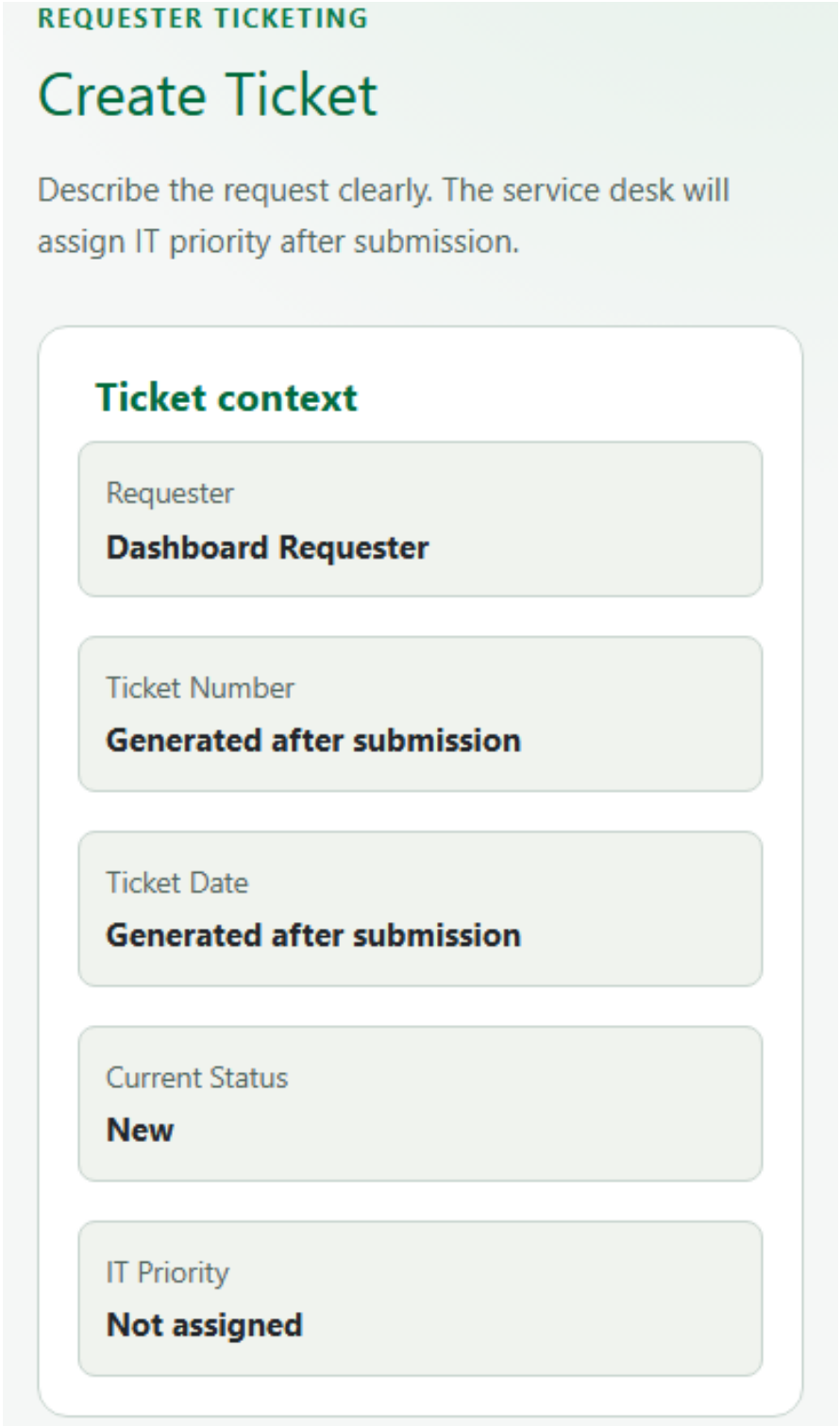
Responsive My Actions - mobile work card

Figure 40 | staff-work-390.png | application 0f169ea. Excerpt: (0, 665, 390, 1020). Original full PNG retained.



Create Ticket - mobile context

Figure 41 | requester-create-390.png | application 0f169ea. Excerpt: (0, 260, 390, 925). Original full PNG retained.



Aligned edit controls - tablet

Figure 42 | action-edit-controls-768.png | application 0f169ea. Unmodified full region capture.

